Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis

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Developmental language disorder affects approximately 7-10% of children in the United States and is associated with long-term difficulties in literacy, academic achievement, and social participation. Language sample analysis (LSA) is widely regarded as a gold-standard approach for evaluating children's language development; yet, it remains underused in clinical and research practice because it depends on labor-intensive and time-consuming manual transcription and coding. Therefore, we propose SATE, a scalable and automated system for child language assessment. It implements a modular machine learning pipeline that reproduces the full LSA workflow, including verbatim transcription, utterance segmentation, speaker identification, maze annotation, lexical and grammatical analyses, and language metrics computation. The pipeline is complemented by a user interface that supports efficient review and correction of model output, allowing researchers and clinicians to validate and refine automated results. Experimental evaluation on child speech corpora demonstrates that SATE substantially outperforms existing baselines, achieving 11.81% word error rate and strong correlations with authenticated results of language metrics. A counterbalanced human-in-the-loop study showed that SATE-assisted LSA reduced task completion time by 55.3% while maintaining comparable accuracy to manual LSA, and usability questionnaires yielded a mean SUS score of 81.25. These results provide evidence that SATE is a practical tool for child language sample analysis.

CCS Concepts: • **Computing methodologies** → **Machine learning approaches**; **Speech recognition**; **Information extraction**; • **Applied computing** → **Health informatics**; • **Human-centered computing** → *User studies*.

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1 INTRODUCTION

Developmental language disorder (DLD) is a neurodevelopmental condition characterized by persistent difficulties in the acquisition and use of language that cannot be explained by hearing loss, intellectual disability, or other biomedical conditions. Large epidemiological studies have estimated prevalence around 7% to 10% of school-age children, underscoring its public health relevance [4, 5, 31, 42]. Children with DLD are at increased risk for reading difficulties, reduced educational attainment, and long-term impacts on employment and mental health [29, 46]. However, only about 30% of children with DLD have been identified [29]. Thus, there is a sustained need for methods that can reliably evaluate children's language development and support both clinical decision making and research on mechanisms and interventions.

Language sample analysis (LSA) occupies a central position in this landscape. Unlike standardized language tests, which typically elicit constrained responses, LSA examines children's spontaneous language use in real-life situations, such as conversation or story telling. This approach enables detailed profiling of children's lexical, morphological, syntactic, and discourse skills, as well as the disfluencies they produce. Research has consistently shown that LSA measures can help clinicians diagnose children with DLD and identify intervention goals [11, 32, 44]. As a result, LSA is widely regarded as a gold-standard method for child language assessment.

Despite the usefulness of LSA, it is still not conducted regularly in clinical settings partly due to the time constraint. For example, a conversational language sample of 15 to 20 minutes is typically needed for a preschool child to obtain the recommended sample length, such as 50 to 100 utterances. Typically, a one-minute language sample takes an experienced SLP 5 to 8 minutes to process. Thus, it may take approximately 1 to 3 hours to process the language sample for a preschooler. Surveys of practicing SLPs report substantial barriers to the routine use of LSA, including the extensive time required for transcription and coding, the need for specialized training, and concerns about scoring consistency [19, 45]. These challenges are further compounded by a persistent shortage of SLPs and their increasing workloads, which limit the feasibility of applying LSA in clinical practice.

A central promise of ubiquitous computing in education is support that reaches learners wherever learning happens, in classrooms, therapy rooms, and homes. For learners facing language challenges, this support remains largely out of reach: most children with DLD are never identified, and for those who are, progress monitoring depends on hours of specialist coding per sample. Diagnosis therefore comes late, and improvement is rarely measured. Removing this analysis bottleneck is a prerequisite for educational environments that can detect, diagnose, and track language difficulties as part of everyday practice.

Existing computational approaches only partially address these limitations. Existing LSA tools such as CLAN [26] and SALT [16] support standardized coding conventions and metric computation, but they generally assume that transcripts and annotations have already been produced by trained human coders. General-purpose and child-adapted ASR systems can reduce part of the transcription burden, but they are not designed to preserve all LSA-relevant phenomena or to produce editable intermediate annotations such as C-unit boundaries, maze labels, speaker labels, and morpheme codes. Existing automated LSA systems also provide limited support for expert verification and correction. Collectively, these gaps highlight the need for a comprehensive, efficient, and verifiable approach to automating Language Sample Analysis while preserving efficient expert oversight.

Therefore, we propose SATE (Speech Annotation and Transcription Enhancer), a human-in-the-loop system for automated child LSA. SATE combines three design choices. First, it implements a domain-adapted pipeline

that converts child speech into structured LSA outputs, including transcript, C-unit boundaries, speaker labels, maze labels, morpheme labels, and language metrics. Second, it introduces end-to-end transcription-and-maze models to examine whether joint modeling can reduce error accumulation between ASR and downstream annotation. Third, it provides an interactive interface that allows experts to review, verify, and correct all intermediate outputs before final metric interpretation. This design positions automation as a support for expert judgment rather than a replacement for expert review.

To validate the proposed system, we evaluate the performance of each machine learning component using objective metrics. We further compare SATE result with those from manual LSA performed by experienced researchers, which demonstrate that SATE produces consistent language metrics and preserves diagnostic insight. In addition, we conduct a counterbalanced human-in-the-loop evaluation, showing that SATE-assisted LSA substantially reduces completion time while maintaining comparable or better final transcript and annotation accuracy than manual LSA.

In summary, our contributions are as follows:

- **A verifiable LSA automation system** that directly addresses a long-standing barrier in speech pathology: the heavy time and expertise burden that has prevented use of LSA. By converting the traditionally manual, low-efficiency workflow into an automated pipeline with interpretable and editable intermediate representations, the system enables reliable LSA at scale.
- **A joint modeling framework** that mitigates error accumulation between transcription and disfluency detection. We introduce two end-to-end architectures and an evaluation protocol that makes annotation performance comparable under ASR discrepancies, demonstrating joint modeling improved robustness to upstream transcription errors.
- **A human-in-the-loop interaction paradigm** that improves the usability of automated LSA, integrating audio-synchronized review with multi-layer visualization and efficient editing of transcripts and linguistic annotations. This is the first system that couples an automated LSA pipeline with an expert-facing correction interface, and we provide preliminary human-in-the-loop evidence of its practical utility through a pilot user study.

2 BACKGROUND AND RELATED WORK

In this section, Section 2.1 first introduces the LSA workflow and the task requirements that motivate the system design. Sections 2.2–2.3 then review related work on children's ASR and component-level automation. Finally, Section 2.4 discusses existing automated LSA systems and the remaining gap for verifiable human-in-the-loop analysis.

2.1 Language Sample Analysis Workflow and Automation Requirements

A typical manual LSA workflow is illustrated in Figure 1. A trained SLP first listens to a child language sample and produces a verbatim orthographic transcript. The transcript is then segmented into communication units (C-units), speaker turns are identified, and disfluencies are annotated as mazes. Inflectional morphemes are subsequently coded at the word level, and summary language metrics are computed. These metrics are then interpreted relative to age expectations, diagnostic needs, and intervention goals.

Several LSA conventions are especially important for automation. A C-unit is commonly defined as an independent clause together with any dependent clauses that belong to it [24]. This definition differs from ordinary sentence segmentation because children's spontaneous speech often contains incomplete clauses, coordination, noncanonical grammar, and disfluencies. Maze annotation identifies repetitions, revisions, and filler words. These segments are clinically meaningful because they may reflect language planning and repair processes, but they are also excluded from some downstream linguistic metrics so that repetitions or abandoned forms do not

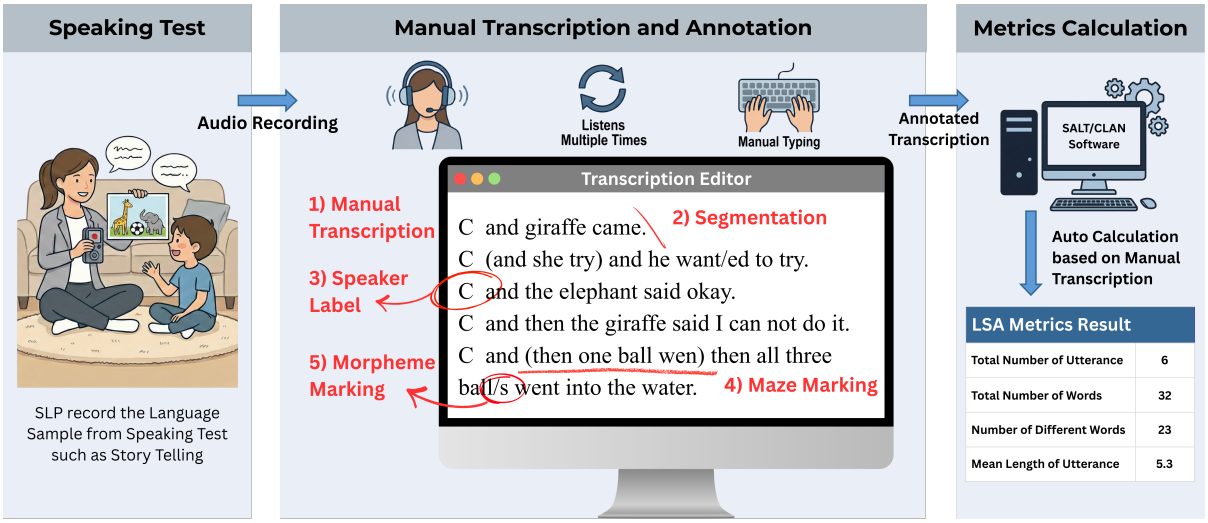


Fig. 1. Typical Language Sample Analysis Workflow.

inflate measures of syntactic or lexical ability. Morphological coding focuses on inflectional morphemes such as plural *-s*, possessive *-s*, third-person singular *-s*, past tense *-ed*, past participle, and present participle *-ing*.

This workflow creates two requirements that are not addressed by ordinary transcription systems. First, the system must preserve verbatim child productions, including disfluencies, grammatical errors, and atypical forms, rather than normalizing them into fluent adult-like sentences. Second, the system must expose intermediate representations for review, because errors in transcription, segmentation, maze annotation, or morpheme coding can change final LSA metrics. These requirements motivate the design of SATE as an automated but verifiable workflow rather than a fully automatic black-box predictor.

2.2 Children's Automatic Speech Recognition and Speech Assessment

Verbatim transcription of child speech is the foundation of automated LSA, yet children's ASR remains a difficult problem. Compared with adult speech, children's speech contains greater acoustic variability, higher pitch, less stable articulation, developing phonology, invented words, frequent disfluencies, and noncanonical grammatical constructions [14]. These properties produce a persistent domain mismatch for ASR models trained primarily on adult fluent speech. Language model priors can further normalize or suppress precisely the atypical productions that LSA needs to preserve, including repetitions, revisions, omissions, and morphological errors [41].

Recent work has therefore explored child-specific adaptation of modern ASR systems. Rolland and Abad studied transfer-based strategies for improving children's ASR, including adaptation methods designed to address the acoustic variability and data scarcity of child speech [36]. Jain et al. examined native and non-native English child speech recognition with Whisper and showed that fine-tuning large pretrained models on child speech can substantially improve performance on child ASR benchmarks [18]. Attia et al. similarly demonstrated that child-oriented ASR adaptation can reduce recognition errors in children's speech [2]. In adjacent educational assessment work, Wang et al. proposed a unified model for oral reading fluency and student prosody, illustrating the broader movement toward task-specific child speech technologies for educational and language assessment contexts [43].

These studies show that child-specific ASR adaptation is essential, but they also leave an important gap for LSA. Most child-ASR work optimizes transcription accuracy or reading-fluency outcomes, whereas LSA requires a transcript that preserves disfluent and clinically meaningful productions and supports downstream C-unit, maze, speaker, and morpheme annotations. SATE builds on the insight that child-domain adaptation is necessary, while extending the objective from transcription alone to a structured and editable LSA workflow.

2.3 Component-Level Automation for LSA Annotations

Speech and language processing research has developed models for transcription, disfluency detection, segmentation, and linguistic analysis. However, these components remain poorly suited for child LSA due to domain mismatch and error propagation.

Disfluency and Maze Detection. Maze annotation is related to disfluency detection in NLP and speech processing. Early text-based methods relied on surface cues such as repeated words [23], while more recent approaches use contextual representations from bidirectional LSTM and Transformer architectures [25, 48]. Audio-based and audio-text methods use prosodic cues and acoustic-textual fusion to improve detection [12, 37, 39]. Recent work has also explored generating disfluency tokens during speech transcription [49]. However, most disfluency systems are designed for adult conversational speech or generic disfluency labels, not for child LSA conventions. In child LSA, maze annotations must be compatible with SALT/CHAT-style coding and must operate reliably on spontaneous language from children with and without DLD.

Applying existing disfluency methods directly to LSA therefore presents three challenges. First, text-only models inherit upstream ASR errors. Second, audio-only models require alignment back to transcript tokens before LSA metrics can be computed. Third, child speech and DLD speech contain disfluency, grammar, and phonological patterns that differ from adult corpora [5]. These challenges motivate a domain-adapted maze classifier and the end-to-end transcription-and-maze models evaluated in this paper.

Utterance and C-unit Segmentation. Sentence segmentation models divide text into sentence-like units for downstream NLP tasks [13, 30]. C-unit segmentation differs from ordinary sentence boundary detection because it follows clinical LSA conventions rather than written punctuation or adult sentence structure. Batchalign includes C-unit-oriented processing for CHAT transcripts [22], but few segmentation systems are trained specifically for children's spontaneous language. Disfluencies, incomplete utterances, and noncanonical grammar are especially common in child speech and can degrade generic segmentation models.

Morphological Analysis. Morphological analysis has long been supported by tools such as CLAN and its MOR program [26]. Neural NLP toolkits such as Stanza provide tokenization, lemmatization, POS tagging, dependency parsing, and Universal Dependencies morphological features [9, 33]. However, LSA morpheme coding differs from generic morphological tagging because it must map linguistic features to clinically meaningful inflectional morpheme categories. In addition, mazes can disrupt the local syntactic context used by morphological analyzers. For this reason, SATE performs morpheme annotation after maze detection and exclusion.

2.4 Automated Language Sample Analysis Systems

Batchalign [22] is a major effort toward automated LSA. It can generate CHAT-format [26] transcripts from speech and connects ASR with downstream transcript processing. For the clinical and research use case targeted by SATE, however, two limitations remain important.

Limitation 1: Sequential pipeline without mitigation of error accumulation. Batchalign integrates existing models into a sequential pipeline where each stage operates independently. Although the authors note that low ASR

error rates minimize downstream impact, the system lacks explicit mechanisms for mitigating error accumulation across stage. The system becomes problematic when applied to challenging populations such as children with DLD, where higher ASR error rates can cascade through segmentation and annotation stages.

Limitation 2: Lack of human-in-the-loop verification. Batchalign outputs finalized transcripts without providing mechanisms for efficient expert review or correction. Clinical practice guidelines require that language assessment results be verified by qualified professionals before they are used to support diagnostic or intervention decisions [1]. Without an efficient and user-friendly interface for human-in-the-loop interaction, the applicability of the system in real-world clinical practice is fundamentally constrained, regardless of its underlying model performance.

In summary, prior work establishes the importance of child-specific ASR adaptation and provides component technologies for disfluency detection, segmentation, and morphological analysis. However, existing approaches do not fully address the combined requirements of child LSA: preserving clinically meaningful verbatim productions, producing structured LSA annotations, reducing error propagation, and supporting expert verification. SATE addresses this gap through domain-adapted models, end-to-end transcription-and-maze architectures, and an interactive interface for efficient human oversight.

3 DATASET AND PREPROCESSING

All training and evaluation data used in this study is from the Edmonton Narrative Norms Instrument (ENNI) dataset [38]. The ENNI corpus was collected to support the assessment of children's narrative language abilities and contains spontaneous speech samples from 361 children aged 4 to 9 years, including both typically developing (TD) children and children diagnosed with DLD. In the dataset, children were prompted to describe short stories composed of 3 sequential pictures from 2 narrative types that introduce linguistic variety.

The ENNI dataset was transcribed in 2 formats. One widely used format is the Codes for the Human Analysis of Transcripts (CHAT), which is publicly available through the TalkBank [26] CHILDES [27] database. CHAT transcriptions encode maze using symbolic annotations (e.g., <, >, [/], or [//]), and each transcription line corresponds to a single C-unit with associated speaker labels and time stamps. For example, a CHAT line such as `*CHI: <he was> [/] he was running.` indicates that the utterance was produced by the child speaker, that the entire line is one C-unit, and that `<he was> [/]` is a repetition maze. Similarly, `*CHI: <he is> [//] he was happy.` represents a revision maze, while speaker prefixes such as `*CHI:` and `*INV:` distinguish child and examiner utterances. As a result, the CHAT format offers reliable reference annotations for evaluating verbatim transcription, maze detection, speaker identification, and C-unit segmentation.

However, the inflectional morpheme annotations are generated automatically by the Computerized Language Analysis (CLAN) software as dependent tiers rather than being manually annotated and verified by experts. Therefore, we did not use the CHAT-format transcripts for evaluating morpheme annotation performance. We incorporated a second version of the ENNI transcripts in Systematic Analysis of Language Transcripts (SALT) format, obtained from prior work [15]. In this dataset, all inflectional morphemes were manually annotated by licensed SLPs, and reliable LSA metrics were computed based on these expert-coded transcripts. The SALT version contains the same language samples as the CHAT version, and its orthographic transcriptions and C-unit boundaries are consistent with the CHAT transcripts. The main difference is that the SALT version includes expert-verified inflectional morpheme coding. For example, SALT marks bound morphemes using suffix annotations appended to word roots, such as `dog/s` for plural *-s*, `walk/ed` for past tense *-ed*, `walk/ing` for present participle *-ing*, and `walk/3s` for third-person singular *-s*. This SALT-format dataset was therefore used for evaluating morpheme annotation and for assessing the overall system performance of SATE.

Table 1. Summary of transcript preprocessing rules for CHAT and SALT formats. "Converted" indicates that the annotation was transformed into structured labels or metadata for model training/evaluation; "Cleaned" indicates that the symbol was removed or normalized from the lexical transcript.

Format	Category	Example	Operation
CHAT	Speaker label	`*CHI:`, `*INV:`	Converted
CHAT	Timestamp for audio boundary	`1223_2058`	Converted
CHAT	C-unit boundary	Each main-tier utterance	Converted
CHAT	Repetition maze	`<he was> [/]`	Converted
CHAT	Revision maze	`<he is> [//]`, `<he is> [/-]`	Converted
CHAT	Filler word	`&-um`, `&=um`, `&~um`, `um`, `uh`, `ah`	Converted
CHAT	Noncompletion	`n(o)t`, `(i)s`	Cleaned
CHAT	Child-invented word marker	`word@c`	Cleaned
CHAT	Pause and other non-lexical label	`(.)`, `(..)`, `(...)`, `&=laughs`	Cleaned
SALT	Inflectional morpheme annotation	`dog/s`, `walk/ed`, `walk/ing`, `walk/3s`	Converted

Before model training and evaluation, CHAT and SALT transcripts were normalized into task-specific structured representations. For CHAT transcripts, speaker labels, timestamps, C-unit boundaries, and maze annotations were extracted from the main-tier utterance lines. Annotations that corresponded to LSA-relevant labels were converted into structured labels, while pause markers, dependent annotations, and other non-lexical markup were removed from the lexical transcript. For SALT transcripts, expert-coded morpheme suffixes were converted into word-level morpheme labels while preserving the same C-unit and lexical transcript structure as the CHAT version. Table 1 summarizes the main preprocessing rules.

For all experiments, the data were split into 70% training and 30% testing sets at the speaker level and the proportions of TD and DLD children were matched across the training and test partitions. Specifically, the split was performed by assigning 70% of speakers to the training set and the remaining 30% of speakers to the test set. No speaker appeared in both partitions. This speaker-level split prevents data leakage caused by having language samples from the same child appear in both training and evaluation, ensuring that test performance reflects generalization to unseen speakers rather than memorization of speaker-specific acoustic or linguistic patterns.

4 SATE: SPEECH ANNOTATION AND TRANSCRIPTION ENHANCER

Our SATE system consists of two mutually reinforcing components as shown in Figure 2. The first component is an automated machine learning pipeline that reproduces the Language Sample Analysis (LSA) workflow, including verbatim transcription, utterance segmentation into communication units, speaker identification, maze annotation, morphological analysis, and computation of language metrics. The second component is an interactive web-based environment that presents model outputs in an interpretable form and enables efficient review, audio-synchronized verification, and correction of model predictions. Together, these components form an integrated system that balances automation with human oversight, ensuring both efficiency and analytic reliability.

This section is organized as follows. Section 4.1 describes our pipeline solution, in which each analytic task is addressed by a individual machine learning model operating in sequence. Section 4.2 presents two end-to-end modeling approaches that unify transcription and annotation within a single model, thereby mitigating the error propagation in sequential pipelines. Finally, Section 4.4 introduces the design of the user interface, including the review panel layout, annotation display, and editing operations that support efficient human-in-the-loop verification and correction.

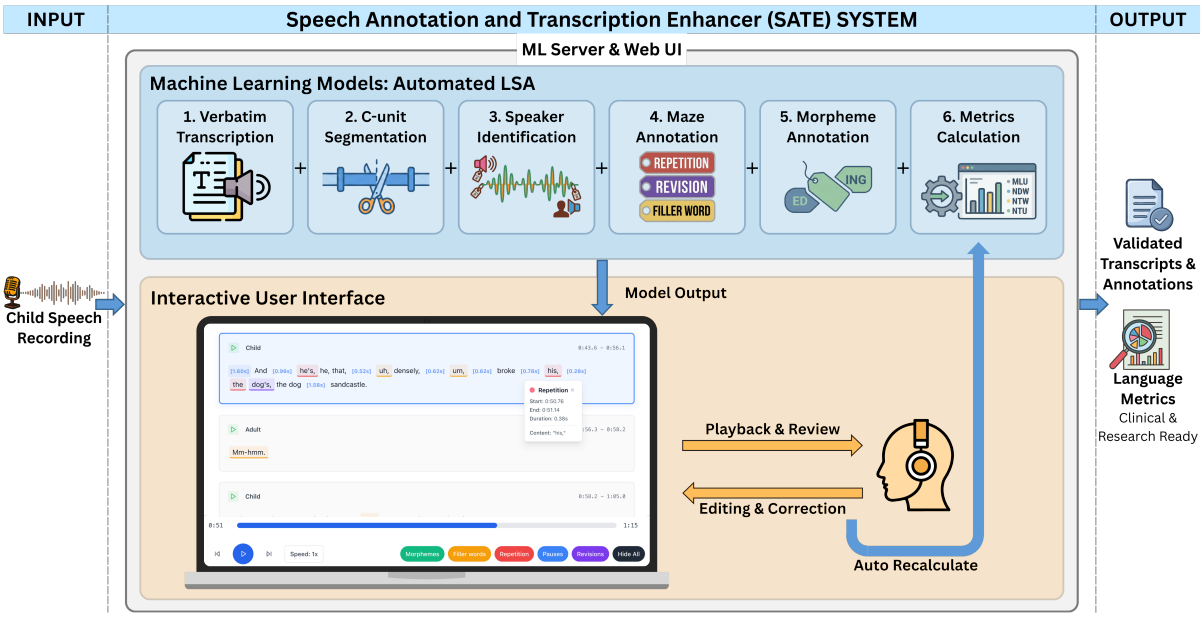


Fig. 2. SATE System Overview. The "Automated LSA" panel summarizes the functional modules supported by SATE and should not be interpreted as a fixed backend execution order. The actual backend configurations are detailed in Figure 3.

4.1 Pipeline Solution of Automatic Language Sample Analysis

In the pipeline solution, SATE implements a sequence of independent models that mirror each step of the LSA workflow. Each component addresses a well-defined subtask and produces structured outputs that serve as inputs to subsequent stages. The upper half of Figure 3 illustrates how raw audio is progressively transformed into LSA metrics through successive model inference steps. This pipeline formulation emphasizes transparency, modularity, and interpretability, and each stage can be independently developed, evaluated and updated.

4.1.1 Verbatim Transcription Model. To develop an ASR system that is accurate for children's speech while preserving everything that is actually spoken in verbatim transcription, SATE adopts a supervised full-parameter fine-tuning strategy using spontaneous child speech dataset that corresponding transcripts preserve speech maze and linguistically relevant deviations, such as morphological errors. This fine-tuning strategy ensures that the ASR model is not only acoustically adapted to children's speech, but also linguistically aligned with the requirements of downstream LSA tasks.

The ASR component is initialized from the publicly released `openai/whisper-small` checkpoint [34]. All encoder and decoder parameters are updated during fine-tuning, with no layers frozen. Audio is resampled to 16 kHz and converted to log-mel features using the backbone's standard preprocessing. Training hyperparameters and decoding settings are listed in Section 4.3.

At inference time, this module takes a complete speech sample audio file as input and produces a continuous verbatim transcript of the entire recording. The output is not segmented at this stage, and maze structures within the transcript are not identified. These analyses are performed by downstream models in the SATE pipeline.

4.1.2 C-units Segmentation Model. After transcription, automated C-unit segmentation in the SATE pipeline operates directly on the textual output and is formulated as a token-level binary classification task, in which

each token is assigned a binary boundary label indicating whether it terminates a C-unit. For an input token sequence $\{w_1, \ldots, w_T\}$, the model predicts $y_t \in \{0, 1\}$, where $y_t = 1$ indicates that token w_t is the final word of a C-unit. We developed a BERT-based token classification model using C-unit boundary labels derived from the public ENNI CHAT transcripts. Because each CHAT main-tier utterance line corresponds to one C-unit, the final word of each line was labeled as a C-unit boundary and all other words were labeled as non-boundaries.

The segmentation model is initialized from `bert-base-uncased` [10] with a linear classification head over contextualized token representations; all transformer layers are fine-tuned. Inputs are truncated or padded to 512 WordPiece tokens. When a word is split into multiple subwords, the boundary label is assigned to the first subword, while remaining subwords and special tokens are masked out of the loss. To use the context window efficiently, adjacent CHAT utterances are concatenated until the resulting sequence approaches the model context limit. Inline CHAT maze symbols are removed, but maze words are retained so the model learns C-unit boundaries in the presence of disfluent child speech. Training hyperparameters are listed in Section 4.3.

4.1.3 Speaker Identification. Language samples often involve interactions between children and adults (e.g., parent, SLP). Thus, distinguishing between speakers is essential for computing metrics for the child specifically, as well as for identifying interaction dynamics. SATE performs speaker identification by combining time alignment with acoustic classification models. Forced alignment is first performed using a wav2vec2-CTC based aligner, which provides temporal localization of each word and yields the start and end time boundaries of each C-unit. These boundaries are then used to segment the full recording into corresponding C-unit audio segments. Each audio segment is passed to the `bookbot/wav2vec2-adult-child-cls` model, an off-the-shelf wav2vec 2.0 audio classification model designed for adult and child speech classification. This two-stage approach enables SATE to classify speakers at the span of each C-unit using acoustic features, improving accuracy while maintaining alignment between utterance-level and speaker-level segmentation.

We do not introduce a novel speaker identification model in this work. Our contribution in this component lies not in model innovation, but in the systematic integration of speaker identification into the automated LSA workflow, ensuring reliable alignment with transcription, segmentation, and subsequent linguistic analyses.

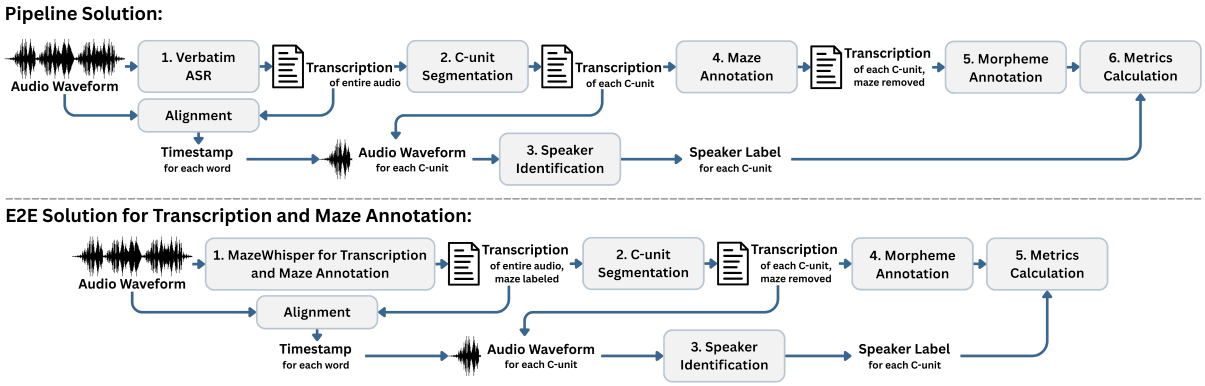


Fig. 3. Comparison of Pipeline and End-to-end Approach for Transcription and Maze Annotation.

4.1.4 Maze Annotation Model. SATE's maze annotation model is also based on a BERT-based token classifier. We formulate maze detection as a token-level multi-label classification problem. Each token receives three independent binary labels indicating whether it participates in a repetition, a revision, or a filled-pause production. Non-maze tokens correspond to the all-zero label vector, so the classifier learns to distinguish ordinary content

words from maze tokens without requiring a separate maze-detection stage. Contextualized embeddings from the BERT transformer backbone are passed through a final linear classification layer and a per-label sigmoid activation function to produce token-level probabilities. Similar to the segmentation model, the maze annotation model is initialized from `bert-base-uncased` [10] with a linear classification head, and all transformer layers are fine-tuned. Training labels are extracted from CHAT maze annotations in the ENNI training split. The lexical transcript is cleaned by removing symbolic CHAT markers while preserving the actual maze words, allowing each word to be paired with its corresponding maze label. Training hyperparameters are listed in Section 4.3.

4.1.5 Morphological Analysis. Rather than developing a new morphological analysis model, our goal is to reliably integrate established linguistic analysis tools into an automated LSA workflow. Accordingly, SATE uses the Stanza natural language processing toolkit [33], which provides universal part-of-speech tags, language-specific POS tags, and rich morphological feature annotations based on the Universal Dependencies framework [9]. Stanza has been validated across a wide range of corpora and languages, making it a stable foundation for morphological analysis.

To align Stanza outputs with the inflectional morpheme categories defined in LSA, we implement a conversion pipeline that maps Universal Dependencies features such as Number, Tense, and Person to the six target morpheme types. This conversion process reproduces the morpheme coding decisions made in manual LSA and yields word-level morpheme labels that can be directly used for metric computation. Importantly, morphological analysis in SATE is performed only after maze has been detected and ignored from the transcriptions. Mazes such as repetitions and revisions often disrupt local syntactic context and can introduce spurious cues for morphological analyzers. By excluding mazes prior to morphological processing, SATE ensures that inflectional features are derived from linguistically well-formed utterances, resulting in more stable and reliable morpheme annotation. Experimental results demonstrate that this design substantially improves the accuracy as shown in Table 6.

4.1.6 Language Sample Analysis Metrics. Once transcription and all processing are complete, SATE computes the LSA metrics widely used in both clinical and research practice. Core measures include the number of total utterances (NTU), number of total words (NTW), number of different words (NDW), and mean length of utterance in morphemes (MLUm). All metric definitions and calculation procedures strictly follow SALT conventions [16], which is the most widely adopted standard among SLPs [45]. In addition to these core measures, SATE also computes lexical diversity metrics, including Type–Token Ratio (TTR) [35] and Vocabulary Diversity (VOC-D) [28], both of which have been extensively validated in LSA research.

The computation of these metrics marks the completion of the LSA workflow. In clinical practice, SLPs typically compare a child's metrics with normative values from age-matched peers to support diagnostic decisions, or track changes in these metrics over time to evaluate the progress of intervention. For researchers, these metrics serve as the foundational starting point for quantitative analyses of children's language development.

4.2 End-to-end Models for Transcription and Annotation

The pipeline solution provides the complete modular SATE workflow and supports all downstream LSA components. However, because transcription and maze annotation are performed by separate models, ASR errors may propagate into maze prediction. We therefore introduce the following E2E models as complementary designs focused specifically on joint transcription and maze annotation. As reported in Table 7, we evaluate maze annotation under three transcription conditions: using fully corrected transcripts, incorporating substitution errors, and incorporating all transcription errors. The results demonstrate a consistent decline in annotation performance as transcription errors increase, illustrating the compounding effect of upstream inaccuracies on downstream tasks. To address this limitation, we adopt an end-to-end modeling strategy that jointly performs

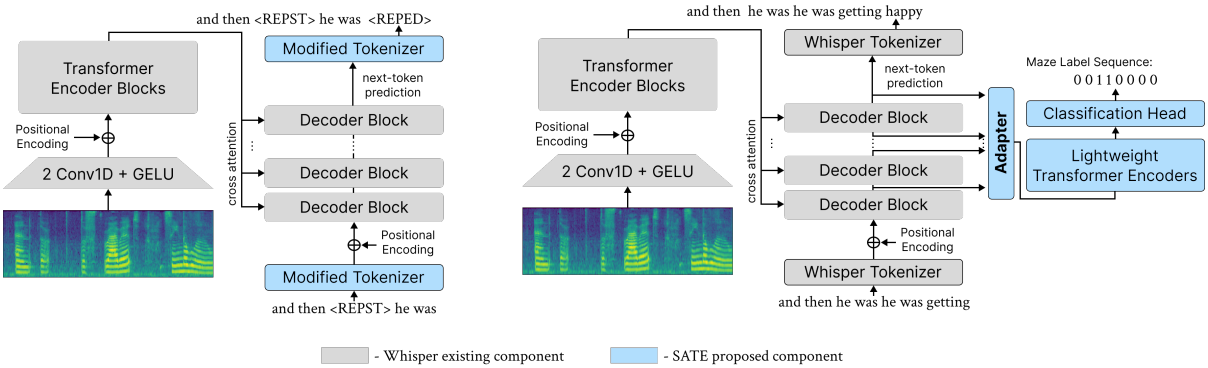


Fig. 4. End-to-end Models for Transcription and Maze Annotation. The Whisper encoder-decoder backbone is reused from `openai/whisper-small`. Newly introduced components are highlighted: tokenizer is modified to add maze special tokens in Solution 1, and the layer-fusion adapter, maze Transformer encoder, and maze classification head in Solution 2.

transcription and maze annotation within a unified optimization framework. By learning both tasks simultaneously, the model reduces error propagation and improves robustness to imperfect intermediate representations.

Specifically, we propose two end-to-end models based on transformer encoder-decoder architectures. The first method treats maze labels as special tokens and directly inserts them into the transcription output. The model is trained to generate both lexical tokens and maze tokens simultaneously. The second approach augments the encoder-decoder model with an additional lightweight transformer encoder that predicts maze labels based on the decoder's hidden representations, separating lexical generation from label prediction while maintaining shared linguistic representations.

In both architectures, transcription and maze annotation are predicted jointly, whereas subsequent processing stages including C-unit segmentation, speaker identification, morphological analysis, and language metric computation follow the same methods as in the pipeline solution. The overall processing flow is illustrated in Figure 3.

4.2.1 End-to-end Solution 1: Inline Label Generation via Special Tokens. Special tokens have been widely used in transformer-based text modeling to encode structural and control information, such as sentence boundary markers, padding tokens, task identifiers, and prompt-conditioning tokens. These tokens allow models to incorporate auxiliary signals into the generation process without altering the core architecture, effectively guiding decoding behavior through learned symbolic representations.

In speech modeling, especially in automatic speech recognition systems, special tokens are adopted to control transcription behavior. For example, Whisper [34] introduces language tokens to specify the target transcription language, as well as voice activity tokens to suppress transcription in non-speech segments. Beyond these uses, special tokens have also been explored for timestamp prediction [17], speaker diarization [20], and sound event detection [47]. Although special-token approaches have been studied in speech and text modeling, their application to disfluency and maze annotation in LSA remains limited. One of the few existing works [49] benchmarks speech disfluency detection using token prediction during transcription. However, that work relies on synthetic training data and does not preserve LSA-compatible lexical transcripts, limiting its applicability for real-world language analysis.

In our model, we design five special tokens to represent maze locations in transcription as shown in Table 2, including the start and end of repetition, the start and end of revision, and a token indicating that the following word is filler word. This design allows explicitly encode the span boundaries and types of maze. We

Table 2. Conversion from original CHAT-format maze annotations to special tokens used in End-to-end Transcription and Maze Annotation Solution 1.

Maze Type	CHAT Original Format	Special Tokens	Transcription Example
Repetition	`and then <he was> [/] he was getting happy and happier.`	`<REPST> <REPED>`	and then <REPST> he was <REPED> he was getting happy and happier.
Revision	`<and then he was so> [//] and then my brother was so happy.`	`<REVST> <REVED>`	<REVST> and then he was so <REVED> and then my brother was so happy.
Filler Word	`the &-um doctor bunny came.`	`<FILLE>`	the <FILLE> um doctor bunny came.

experimented with alternative designs, such as inserting a special token before each word, but found that excessive insertion of special tokens degraded transcription quality by overloading the decoding space. The adopted token design therefore balances annotation expressiveness with transcription accuracy. To enable the model to generate special tokens during transcription, we introduce two key modifications. First, we insert special tokens into training transcripts by transforming symbolic maze annotations from CHAT-format corpora. Second, we extend the tokenizer vocabulary to include the special tokens, allowing the decoder to model them as part of the output space. These modifications are implemented on top of the Whisper encoder–decoder architecture, as illustrated in Figure 4.

E2E Solution 1 keeps the original Whisper backbone unchanged and modifies only the target token space used by the decoder. Specifically, we start from the same `openai/whisper-small` checkpoint used in the pipeline ASR model and extend the Whisper tokenizer with five maze tokens: <REPST>, <REPED>, <REVST>, <REVED>, and <FILLE>. After adding these tokens, the decoder token embedding matrix and output vocabulary projection are resized so that the new symbols can be generated in the same way as ordinary transcription tokens. These maze tokens are constructed by converting the original CHAT maze annotations shown in Table 2. During the training stage, all Whisper parameters are fine-tuned, and the shared training hyperparameters and decoding settings are reported in Section 4.3.

Through this design, the model learns to produce lexical transcription and maze annotation simultaneously from raw speech input. Compared with the pipeline solution, this joint modeling approach introduces only a negligible number of additional parameters and reduces error accumulation by optimizing both tasks within a single unified framework. Importantly, speech mazes are expressed through both lexical and acoustic cues. In the model architecture, cross-attention integrates acoustic representations from the audio encoder with the previously generated text tokens. This design enables effective multimodal fusion during decoding, allowing the model to jointly leverage complementary acoustic and textual information, thereby improving the robustness and accuracy of maze detection.

Despite these advantages, jointly predicting lexical and maze tokens imposes a higher modeling burden on the decoder. Under fixed model capacity, this results in a slight degradation in transcription accuracy compared with models trained to predict text tokens alone, as shown in Table 4. To address this limitation, we propose a second end-to-end architecture described in the following section.

4.2.2 End-to-end Solution 2: Decoder-Augmented Token-Level Labeling. Our second end-to-end architecture is built upon the Whisper transformer encoder-decoder backbone, but decouples lexical generation from maze labeling to reduce the modeling burden. The model generates standard transcription tokens as in conventional

ASR, while maze labels are predicted by an additional branch that reuses the decoder hidden representations. In contrast to Solution 1, this model does not modify the Whisper tokenizer or transcription vocabulary. The ASR pathway therefore remains the original Whisper transcription model, while the newly introduced components are the layer-fusion adapter and the maze-labeling branch attached to the decoder representations. Specifically, we design the following three components to achieve that:

Attention-based Adapter for Layer Fusion. Let $\mathbf{H}^{(l)} \in \mathbb{R}^{T\times d}$ denote the decoder hidden states at layer $l \in \{1,\dots,L\}$ for a token sequence of length T. We first inject a learnable layer positional embedding $\mathbf{P}^{(l)} \in \mathbb{R}^d$ to distinguish representations from different decoder depths:

$$\tilde{\mathbf{H}}^{(l)} = \mathbf{H}^{(l)} + \mathbf{P}^{(l)}. \tag{1}$$

For each token position $t \in \{1,\dots,T\}$, we use the hidden representation from the final decoder layer as the query and attend over the stacked representations from all decoder layers. The attention weights are computed independently for each token and normalized across the layer dimension. As a result, the model assigns a layer-wise attention weight $\alpha_{t,l}$ to each decoder layer for a given token position, indicating how much each decoder layer contributes to the representation used for maze labeling.

$$\alpha_{t,l} = \mathrm{softmax}_l\left(\frac{\left(\mathbf{W}_q\mathbf{h}_t^{(L)}\right)^\top\left(\mathbf{W}_k\tilde{\mathbf{h}}_t^{(l)}\right)}{\sqrt{d}}\right). \tag{2}$$

Given these weights, we compute a fused token representation by taking a weighted sum of the value projections from all decoder layers:

$$\mathbf{z}_t = \sum_{l=1}^{L}\alpha_{t,l}\,(\mathbf{W}_v\tilde{\mathbf{h}}_t^{(l)}). \tag{3}$$

Here, $\mathbf{W}_v\tilde{\mathbf{h}}_t^{(l)} \in \mathbb{R}^d$ denotes the value representation of token t at decoder layer l, and $\mathbf{z}_t \in \mathbb{R}^d$ represents a layer-fused embedding that combines information from multiple decoder depths. Intuitively, this aggregation allows the model to selectively emphasize lower-layer representations that capture local lexical or acoustic-aligned cues, or higher-layer representations that encode more abstract linguistic context, depending on which cues are most informative for maze detection at that token.

Then, the fused representation $\mathbf{z}_t$ is passed through an output projection:

$$\mathbf{m}_t = \mathbf{W}_o\mathbf{z}_t. \tag{4}$$

The resulting vector $\mathbf{m}_t \in \mathbb{R}^{d_m}$ serves as the input to the subsequent Transformer encoder. While $\mathbf{z}_t$ aggregates information across decoder layers, $\mathbf{m}_t$ maps this fused representation into a feature space that is optimized for downstream maze classification.

Maze Transformer Encoder and Label Prediction. The fused sequence $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_T] \in \mathbb{R}^{T\times d_m}$ is then processed by a lightweight transformer encoder with four layers. This maze transformer encoder uses hidden dimension $d_m = 768$, 12 attention heads, and dropout 0.1. The encoder output is fed into a classification head to produce logits $\mathbf{O} \in \mathbb{R}^{T\times C}$, where $C = 4$ corresponds to fluent, repetition, revision, and filler words labels.

Joint Optimization Objective. The model is jointly optimized for transcription and maze labeling with a weighted multi-task objective:

$$\mathcal{L} = (1-\lambda)\,\mathcal{L}_{\mathrm{ASR}} + \lambda\,\mathcal{L}_{\mathrm{maze}}. \tag{5}$$

$\mathcal{L}_{\mathrm{ASR}}$ is standard cross entropy over the transcription vocabulary. $\mathcal{L}_{\mathrm{maze}}$ uses focal loss [21] to address class imbalance, since maze tokens account for a small fraction of all tokens. The Whisper backbone is initialized from

Table 3. Training settings for the learnable SATE components. Effective batch size is computed as per-device batch size multiplied by gradient accumulation steps.

Component	Base Model	Loss	LR	Warmup	Eff. Batch	Budget / Selection
ASR (Pipeline)	`openai/whisper-small`	CE	1×10^{-5}	500 steps	16 (8×2)	10,000 steps / min val WER
C-unit Segmentation	`bert-base-uncased`	CE	2×10^{-5}	10% steps	8	30 epochs / max val F1
Maze (Pipeline)	`bert-base-uncased`	BCE	3×10^{-5}	10% steps	16	10,000 steps / max val micro-F1
E2E Solution 1	`openai/whisper-small`	CE	1×10^{-5}	500 steps	16 (8×2)	10,000 steps / min val WER
E2E Solution 2	`openai/whisper-small`	CE + focal	1×10^{-5}	500 steps	16 (8×2)	10,000 steps / min val WER

the same `openai/whisper-small` checkpoint as Solution 1 and the pipeline ASR model. The adapter parameters, the four-layer maze transformer encoder, and the maze classification head are initialized from scratch. All backbone, adapter, and maze-branch parameters are optimized jointly, with no frozen Whisper layers.

Training audio, transcript targets, and parallel word-level maze labels are derived from the ENNI training set using the same CHAT preprocessing described at Section 3. Unlike Solution 1, which rewrites maze annotations as inline special tokens, Solution 2 keeps the transcript target as a standard Whisper transcription sequence and represents maze annotations as a parallel label sequence. The auxiliary classifier is therefore supervised at token positions aligned with the output transcript tokens. The multi-task weight $\lambda = 0.5$, focal-loss settings, and other shared training hyperparameters are reported in Section 4.3.

At inference time, the ASR backbone first generates a clean transcript using greedy decoding. The generated transcript is then re-tokenized and passed back through the joint model together with the audio features to obtain token-level maze logits. Maze predictions are selected by argmax and aggregated to word level by majority vote across the subword pieces of each predicted word.

Overall, this decoder-augmented architecture with joint optimization encourages decoder representations to remain reliable for transcription while being discriminative for maze detection.

4.3 Implementation Details

To support reproducibility, we summarize the main training settings for each learnable component in Table 3. All models were trained on the ENNI training partition described in Section 3. The training partition was further split into 90% training and 10% validation data for checkpoint selection. Unless otherwise stated, models were optimized with AdamW, linear warmup followed by linear decay, gradient clipping with maximum norm 1.0, and mixed-precision training when supported. A fixed random seed of 42 was used for ASR and end-to-end model training. No layers were frozen for the fine-tuned ASR, BERT-based, or end-to-end models. The speaker identification model, morpheme annotation were used off the shelf and was not fine-tuned.

For the pipeline maze classifier, binary cross-entropy was applied independently to repetition, revision, and filled-pause labels, with a threshold of 0.5 at inference. For E2E Solution 2, the multi-task objective was $\mathcal{L} = 0.5\,\mathcal{L}_{\text{ASR}} + 0.5\,\mathcal{L}_{\text{maze}}$, where $\mathcal{L}_{\text{maze}}$ used focal loss with $\gamma = 2.0$ and uniform $\alpha = 1.0$. During inference, Whisper-based models used greedy decoding with beam size 1, temperature 0.0, length penalty 1.0, and timestamp prediction disabled. E2E Solution 1 additionally disabled forced decoder prompts and token suppression to allow generation of the added maze tokens.

4.4 User Interface for Efficient Review and Editing

Although automation is essential for efficiency, human verification remains a necessary component to ensure that the final results are trustworthy. In the SATE workflow, users begin by uploading audio recordings to the server, where a deployed machine learning pipeline automatically processes it and returns structured results to

the front-end interface. These results include C-unit segmented transcripts, speaker labels, word-level maze and inflectional morpheme annotations, and computed LSA metrics. Users can then play back the audio within the interface, inspect the transcription and annotations, and correct any errors generated by the model. Through this review and editing process, all intermediate LSA result are validated, ensuring that the final metric reports are reliable and suitable for clinical or research use.

Prior work shows that usability, cognitive load, and interface transparency strongly affect experts' ability to efficiently and reliably validate system outputs, particularly in multimodal transcription and annotation tasks involving both audio and text [8, 40]. Guided by these principles, SATE incorporates an audio-synchronized and annotation-aware interface. The design focuses on two objectives: presenting transcriptions and maze annotations in a manner that is transparent, to support rapid inspection and comprehension, and to enable efficient editing operations that allow users to correct errors with minimal cognitive load. These two design emphases are addressed in the following sections.

4.4.1 Review Panel Design. The review panel is organized into three coordinated regions: a navigation panel on the left, a transcription panel in the center and a language analysis panel on the right, as illustrated in Figure 5. The central transcription panel consists of vertically stacked units corresponding to individual C-units. Each unit displays the speaker label, the orthographic transcript and any annotations such as maze or inflectional morpheme. During audio playback, the active C-unit and word are highlighted in synchronize, which significantly reduces search time and facilitates the human detection of transcription or annotation errors.

Within each C-unit, maze and morpheme annotations are visually distinguished using different highlight styles, enabling users to recognize annotation types at a glance while preserving readability of the transcript. Additional details remain accessible: when a user selects an annotated token, the corresponding metadata, such as maze category or morpheme suffix, is popped up on demand, supporting closer inspection without cluttering the primary view. The right side analysis panel include LSA metrics computed from the current transcript and selected speaker, updating dynamically in response to user edits.

4.4.2 Editing Operations. To support efficient and reliable human-in-the-loop correction, SATE provides two types of editing operations that align with the hierarchical structure of LSA transcription: transcription and annotation editing and unit editing, as illustrated in Figure 5.

Transcription and annotation editing enables revision of lexical content and maze, morpheme labels. When a user double-clicks a C-unit, the interface transitions into an editable text in SALT annotation format, which uses symbolic markers to encode annotations. Users can correct transcription errors, adjust maze boundaries or revise morphological markers. Unit editing supports modifications including splitting a C-unit, merging adjacent units, reassigning speaker labels and adding, excluding or deleting units.

The combined design of editing functions enables users to validate and correct all intermediate output underlying LSA conventions, ensuring that the final metrics results reflect expert judgment while retaining the efficiency benefits of automation. More importantly, this design transforms SATE from an automated LSA model into a fully usable tool for clinical and research practice.

5 EXPERIMENTS

This section presents a comprehensive evaluation of the proposed SATE system, ranging from component-level model performance to system-level and human-in-the-loop validation. First, we evaluate each module in the pipeline solution independently, including automatic speech recognition, C-unit segmentation, maze annotation, and morphological analysis. Second, we evaluate the proposed end-to-end models that jointly perform transcription and maze annotation, and compare their results with the pipeline solution to examine robustness

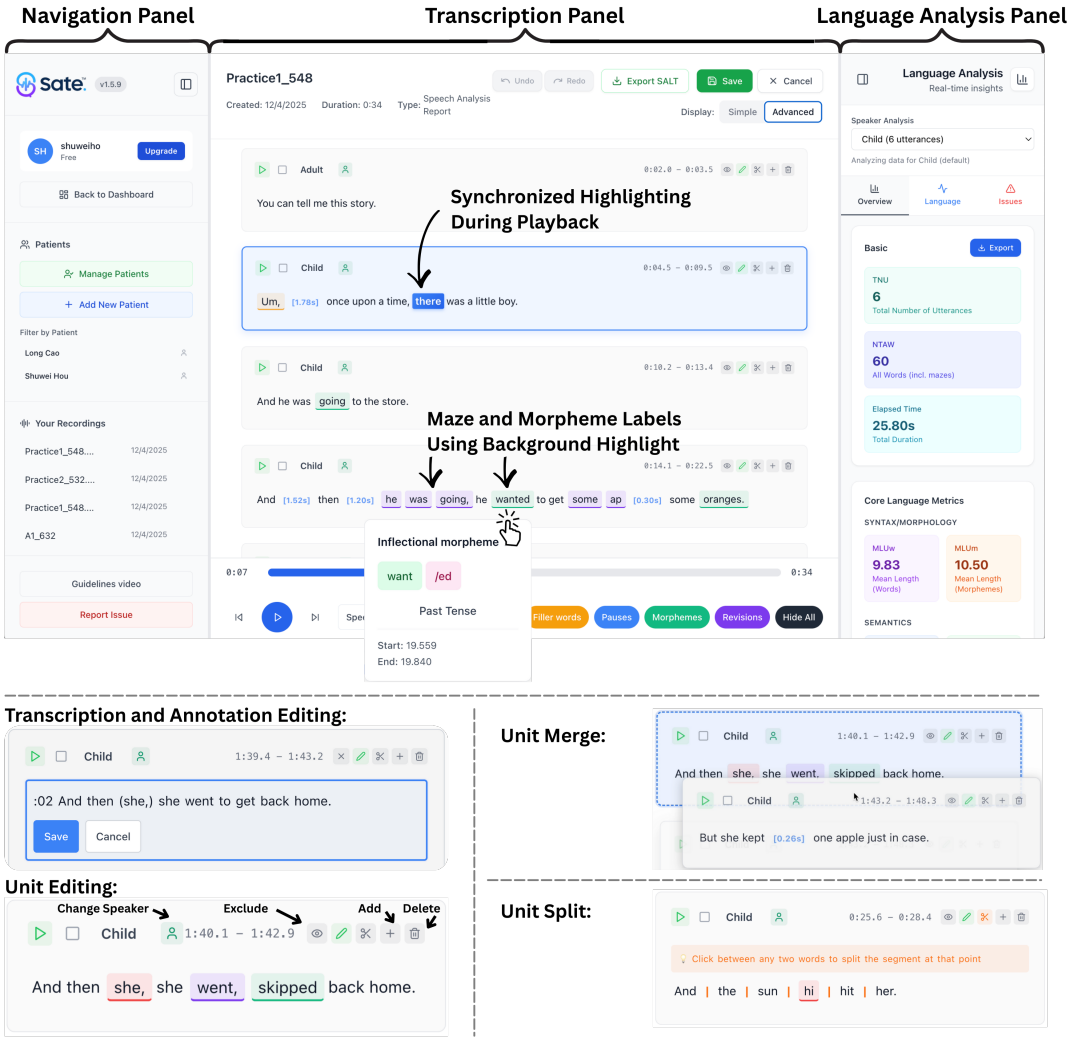


Fig. 5. User Interface of SATE System.

under upstream ASR errors. Third, we assess the overall performance of SATE by comparing automatically generated LSA metrics with those produced by expert-coded reference transcripts, without any manual review or correction, to examine the consistency between automated and human analysis. Finally, we report a counterbalanced human-in-the-loop evaluation that compares SATE-assisted LSA with manual LSA in terms of task completion time, final transcript and annotation accuracy, editing behavior, and user-perceived usability.

5.1 Evaluation of Pipeline Solution

5.1.1 Automatic Speech Recognition. Word error rate (WER) is used to evaluate the performance of ASR component of SATE pipeline solution. WER is defined as the sum of substitution, deletion, and insertion errors divided by the total number of reference words, and is the standard metric used across ASR research. As shown in Table

Table 4. Practical comparison against existing systems without task-specific adaptation. We evaluate ASR performance on two dimensions: (1) maze-word transcription, where evaluation is restricted to words annotated as maze words in the ground truth, and (2) overall transcription on all words. Insertion errors are reported only for all words: an inserted word exists only in the ASR output, whereas maze status is only exists in the expert annotation of the reference transcript when we evaluate ASR performance only, so insertions cannot be attributed to the maze-word subset. Only our models are explicitly adapted to the child-speech story-retelling domain of ENNI, underscoring the necessity of domain-specific fine-tuning for this task.

Type	Model	Maze Words (12.23%)			All Words			
		WER	Substitution	Deletion	WER	Substitution	Deletion	Insertion
General ASR	Whisper Small	57.00%	17.03%	39.97%	31.79%	14.77%	12.43%	4.59%
	Whisper Medium	56.17%	13.27%	42.90%	28.33%	11.82%	12.35%	4.16%
	Whisper Large v3	58.59%	11.15%	47.44%	25.63%	9.80%	12.73%	3.10%
	Batchalign	43.76%	15.92%	27.84%	23.75%	11.25%	8.09%	4.41%
	ElevenLabs Scribe v1	39.27%	15.83%	23.44%	27.61%	6.37%	18.64%	2.61%
	ElevenLabs Scribe v2	38.79%	13.62%	25.17%	27.30%	**5.63%**	19.65%	**2.02%**
Child-speech adapted	ChildASR 1 [18] (Whisper Large v2)	41.83%	19.01%	22.82%	39.81%	13.70%	7.78%	18.34%
	ChildASR 2 [2] (Whisper Medium)	37.96%	14.76%	23.20%	26.58%	10.94%	8.73%	6.91%
Ours (ENNI-adapted)	**Ours (Pipeline)**	22.81%	**13.24%**	9.57%	**11.81%**	6.92%	2.73%	2.16%
	Ours (E2E 1)	**22.51%**	14.88%	**7.63%**	12.13%	7.17%	**2.64%**	2.32%
	Ours (E2E 2)	22.76%	13.62%	9.13%	12.01%	6.98%	2.70%	2.33%

Table 5. C-unit Segmentation Model Performance. Batchalign and SaT were evaluated using their released/default models without ENNI-specific fine-tuning; our model was fine-tuned on the ENNI training split.

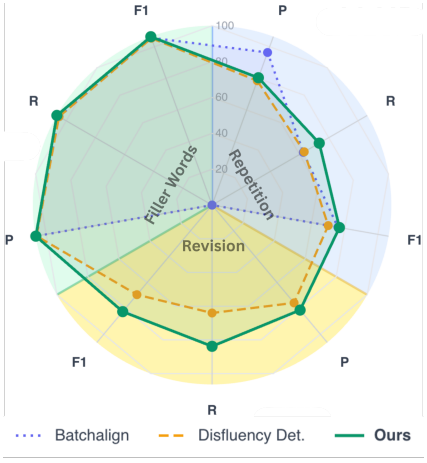
Model	Precision	Recall	F1
SaT 3L [13]	74.2	47.7	58.1
Batchalign [22]	84.3	93.0	88.4
Ours	**93.9**	**94.3**	**94.1**

4, we compare three categories of models: general ASR models with official pretrained weights, child-speech-adapted models from prior work fine-tuned on non-ENNI child speech datasets, and our ENNI-adapted models fine-tuned on the ENNI training set. This comparison should be interpreted as a practical deployment benchmark rather than a controlled comparison of ASR architectures, because the external systems were evaluated using their released models without ENNI-specific fine-tuning.

This evaluation establishes the transcription module as a reliable component for downstream analysis and demonstrates that domain-adapted models are essential for consistent performance in child language research.

5.1.2 C-unit Segmentation. C-unit segmentation is formulated as a token-level binary classification task in which each token is predicted as either a boundary word or a non-boundary word. To isolate segmentation performance from upstream ASR errors, we used ground-truth transcripts as input and compared our segmentation model with two baselines: the segmentation method in Batchalign [22] and the SaT [13] model, as shown in Table 5. Both baselines were evaluated using their released/default models without additional fine-tuning on the ENNI training split. Therefore, this comparison should be interpreted as a practical benchmark of existing segmentation tools on child LSA data rather than a controlled architecture comparison under identical training conditions.

5.1.3 Maze Annotation. Maze annotation performance was evaluated across three categories: repetitions, revisions, and filler words. The evaluation used ground-truth transcripts that were segmented into C-units as input to eliminate the influence of upstream ASR and segmentation errors and enable direct evaluation of maze annotation performance. We compared our model against Batchalign and another baseline disfluency detection model [37]. The baseline systems were evaluated using their released/default models or settings without additional fine-tuning on the ENNI training split, whereas our BERT-based maze classifier was fine-tuned on the ENNI training data. Thus, this comparison evaluates how existing disfluency or LSA annotation tools perform when directly applied to child LSA transcripts, while also showing the benefit of domain-specific adaptation. Results in Figure 6 show that our fine-tuned BERT-based classifier achieves higher accuracy across all maze categories, demonstrating its effectiveness in children's language sample analysis.



Maze Type	Metric	Batchalign [22]	Disfluency [37]	Ours
Repetition	Precision	**90.55**	73.77	75.56
	Recall	58.80	59.41	**69.06**
	F1	71.30	65.81	**72.17**
Revision	Precision	N/A	71.19	**76.42**
	Recall	N/A	60.18	**78.76**
	F1	N/A	65.22	**77.57**
Filler	Precision	**100**	99.44	99.86
	Recall	98.61	98.61	**100**
	F1	99.30	99.02	**99.93**

Fig. 6. Visualization and comparison of maze annotation performance across Batchalign [22], the disfluency detection method from [37], and our model. Baseline systems were evaluated without ENNI-specific fine-tuning; our model was fine-tuned on the ENNI training split.

5.1.4 Morpheme Annotation. For this evaluation, the model was also provided with clean ground-truth transcripts that had already been correctly segmented into C-units. Results reported in Table 6 demonstrate that the Stanza-based morpheme annotation module performs exceptionally well, accurately mapping linguistic morphological features to the inflectional morpheme categories defined in LSA.

In addition, we examined the impact of maze structures on morpheme detection accuracy by comparing model performance on transcripts that retained maze content with performance on transcripts in which maze were removed prior to analysis. As shown in Table 6, morpheme detection performance improved when maze structures were excluded. This result reflects the fact that maze phenomena often disrupt grammatical structure, which can obscure morphological cues and negatively affect downstream linguistic analysis. Therefore, we identified and excluded the maze before morpheme detection in the SATE pipeline, to generate more stable and reliable morpheme annotation results.

5.1.5 Evaluation of Morpheme Annotation on Different ASR Transcriptions. The preceding evaluation isolates morpheme annotation performance using clean reference transcripts. However, in the complete SATE workflow, the morpheme annotation module operates on automatically generated ASR transcripts. Therefore, we further

Table 6. Morpheme annotation performance on clean reference transcripts. Because the Stanza-based morpheme annotation module and the rule-based mapping procedure were used off the shelf without ENNI-specific training or fine-tuning, this evaluation was conducted on the full expert-coded ENNI/SALT dataset rather than only the held-out test split.

	Maze Included			Maze Removed		
Morpheme Type	Precision	Recall	F1	Precision	Recall	F1
Plural	93.98	95.25	94.61	94.15	96.60	95.36
Possessive	98.22	84.87	91.06	98.56	87.69	92.81
3rd Person Singular	88.44	92.63	90.49	88.58	95.08	91.71
Past Tense	89.94	81.19	85.34	90.04	83.23	86.50
Progressive	98.90	78.40	87.46	98.90	79.86	88.37
Overall (Macro Avg.)	93.90	86.47	89.79	**94.04**	**88.49**	**90.95**

evaluated how upstream transcription quality affects downstream morpheme annotation by applying the same fixed morpheme annotation procedure to outputs from different ASR systems.

Because ASR outputs and reference transcripts do not necessarily contain identical word sequences, morpheme annotation was evaluated after aligning the ground-truth and ASR word sequences with Levenshtein alignment, using equal costs for substitution, insertion, and deletion. A ground-truth morpheme predicted with the correct type at its aligned ASR word is a true positive (TP). A ground-truth morpheme whose aligned word lacks the correct label is an aligned false negative (FN_{aligned}, a detector error), and a ground-truth morpheme whose word was deleted by the ASR is a misaligned false negative ($FN_{\text{misaligned}}$, an ASR-induced error). Symmetrically, remaining predictions on aligned words are aligned false positives (FP_{aligned}), and predictions on ASR-inserted words are misaligned false positives ($FP_{\text{misaligned}}$).

Figure 13 in Appendix C visualizes this decomposition. The number of ground-truth morphemes is fixed across ASR systems, so the larger $FN_{\text{misaligned}}$ segments of error-prone ASR systems directly reflect morphemes that become unrecoverable when the corresponding words are missing from the transcript, while the prediction-side panels separate detector over-prediction from predictions on inserted words. The decomposition thus makes explicit how much of the downstream error is attributable to the morpheme detector versus upstream transcription.

We evaluated each ASR condition under two settings: in the *with-maze* setting, the raw ASR transcript was used directly, whereas in the *maze-removed* setting, words predicted as mazes were removed before morpheme evaluation, using the external BERT-based maze classifier for the non-E2E systems and each model's own maze annotations for Ours-E2E-1 and Ours-E2E-2. This comparison tests whether excluding disfluent material improves morpheme annotation under realistic transcription conditions, as suggested by the clean-transcript results in Table 6.

For each morpheme category, we computed total precision, recall, and F1 by including both detector-induced and ASR-induced errors:

$$P_{\text{total}} = \frac{TP}{TP + FP_{\text{aligned}} + FP_{\text{misaligned}}}, \quad R_{\text{total}} = \frac{TP}{TP + FN_{\text{aligned}} + FN_{\text{misaligned}}}, \tag{6}$$

$$F1_{\text{total}} = \frac{2P_{\text{total}}R_{\text{total}}}{P_{\text{total}} + R_{\text{total}}}. \tag{7}$$

Figure 7 summarizes the downstream morpheme annotation results. The ASR models developed in this work achieve the strongest downstream morpheme performance in both macro and micro F1, outperforming both

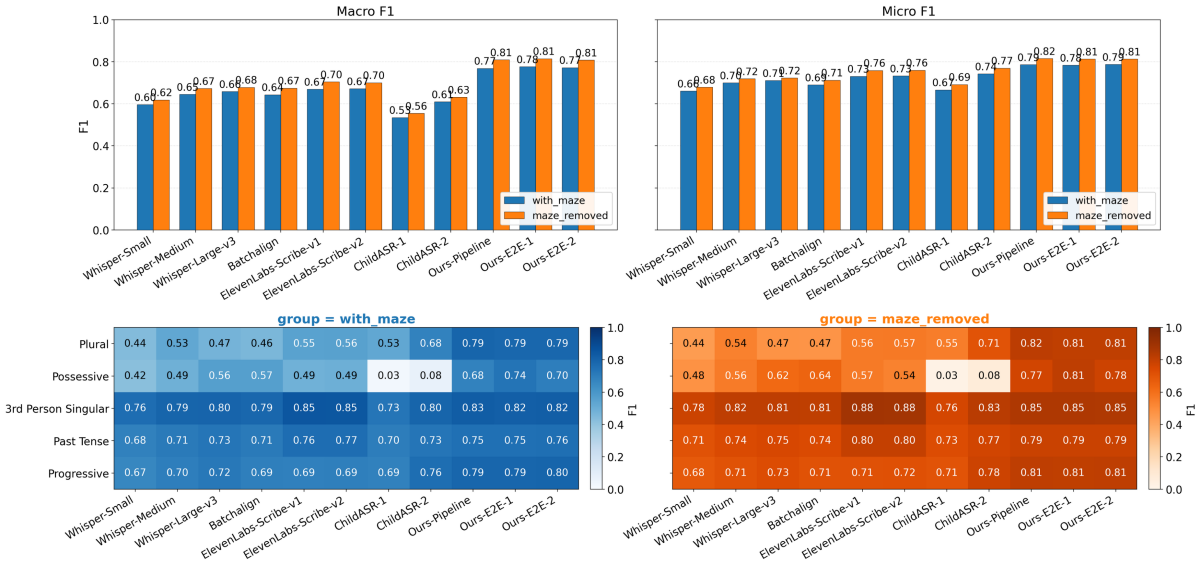


Fig. 7. Downstream morpheme annotation performance across ASR transcriptions. The top row reports macro and micro F1 for each ASR system under the with-maze and maze-removed settings. The bottom row reports per-morpheme F1 scores for five morphemes.

general-purpose ASR systems and child-speech baselines; the benefit of our ASR adaptation is therefore not limited to lower WER, as improved child-speech transcription translates directly into more reliable downstream linguistic annotation. Maze removal yields a further consistent improvement for most ASR systems, supporting the design choice in SATE to exclude mazes before morpheme annotation. Across morpheme categories, possessive morphemes remain the most difficult.

5.2 Evaluation of End-to-end Transcription and Maze Annotation Model

Transcription performance of the two end-to-end models is reported in Table 4: both achieve WER comparable to the pipeline ASR model, indicating that joint optimization preserves recognition accuracy.

Evaluating maze annotation is more challenging because the predicted and reference transcripts do not share identical word sequences. We therefore align them with the same minimum-edit-distance procedure as in Section 5.1.5: correctly transcribed and substituted tokens retain a one-to-one label correspondence, whereas deletion errors introduce false negatives and insertion errors introduce false positives. Annotation performance is then evaluated under three nested conditions, as reported in Table 7: correctly transcribed tokens only (C), additionally including substitution errors ($C + S$), and including all ASR errors ($C + S + D + I$).

Although the end-to-end models do not consistently outperform the pipeline solution in absolute F1, they are more robust to transcription errors. Under the correctly-transcribed condition, the pipeline achieves the highest F1 for all three maze types; as the evaluation scope expands, the best-performing method shifts, and under the all-errors condition E2E Solution 2 leads for repetition labeling and E2E Solution 1 for filler words. To quantify this robustness, we measure how much the F1 score of method m for maze category k decreases when the evaluation

scope expands from C to $E \in \{C + S,\ C + S + D + I\}$:

$$\Delta_{\text{rel}}(m,k,E) = \frac{F^C_{m,k} - F^E_{m,k}}{F^C_{m,k}} \times 100\%, \qquad \Delta_{\text{abs}}(m,k,E) = F^C_{m,k} - F^E_{m,k}. \tag{8}$$

Here, $F^C_{m,k}$ and $F^E_{m,k}$ denote the corresponding F1 scores, normalized to $[0,1]$ when computing the absolute decrease. Each cell of Table 8 reports Δ_{rel}, with Δ_{abs} in parentheses. The end-to-end models exhibit smaller degradation than the pipeline under both expanded conditions, indicating that joint modeling mitigates the accumulation of upstream transcription errors.

Table 7. Maze Annotation Performance across Different Transcription Error Scopes and Solutions. The three evaluation conditions are: Correctly Transcribed Tokens Only, which evaluates only tokens correctly recognized by ASR; Correctly Transcribed Tokens + Substitution Errors, which additionally includes aligned substitution errors; and All ASR Errors Included, which further accounts for insertion- and deletion-induced false positives and false negatives. Bold values indicate the numerically best value among the three methods for each metric within a maze type; bolding does not indicate statistical significance.

	Pipeline Solution			E2E Solution 1			E2E Solution 2		
Maze Type	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Condition: Correctly Transcribed Tokens Only									
Repetition	**73.80**	69.25	**71.45**	69.18	65.43	67.25	70.19	**71.87**	71.02
Revision	**72.51**	72.96	**72.73**	68.47	71.16	69.76	61.66	**75.37**	67.83
Filler Words	**100**	**100**	**100**	100	99.82	99.91	99.01	88.32	93.36
Condition: Correctly Transcribed Tokens + Substitution Errors									
Repetition	**68.73**	62.39	65.41	65.17	61.20	63.12	66.32	**65.79**	**66.05**
Revision	**68.88**	69.88	**69.38**	65.85	69.64	67.69	60.54	**72.63**	66.03
Filler Words	85.59	**88.79**	**87.16**	86.31	87.84	84.51	**89.34**	78.25	83.43
Condition: All ASR Errors Included									
Repetition	**65.07**	54.83	59.51	59.96	54.68	57.20	62.61	**57.73**	**60.07**
Revision	**64.09**	62.86	**63.47**	61.61	62.61	62.11	56.26	**65.87**	60.69
Filler Words	80.51	79.39	79.94	80.67	**80.78**	**80.72**	**82.95**	71.17	76.61

5.3 Evaluation of the Overall Performance of SATE

While modular evaluation provides insight into individual components, a system-level assessment is necessary to evaluate SATE as a complete LSA system that converts raw child speech into clinically interpretable language metrics. To this end, we computed the most widely used LSA metrics (NTU, NTW, NDW, and MLUm) from the transcripts and annotations generated by each SATE variant, and from Batchalign as a complete-system automated baseline, and compared them against the metrics derived by SLP experts. Agreement is reported as Pearson correlation, mean absolute error, mean signed difference, and standard deviation of differences in Table 9, complemented by Bland-Altman analyses [6] in Figure 8 and statistical tests of systematic bias and residual variance in Table 10.

Table 8. Error Accumulation Comparison. Each cell reports the relative F1 decrease Δ_{rel} defined in Equation (8) when moving from the correctly transcribed condition to a broader evaluation condition that includes ASR-induced errors; values in parentheses give the corresponding absolute decrease Δ_{abs}. Smaller values indicate less performance degradation and therefore stronger robustness to transcription-error accumulation. Bold values indicate the smallest degradation within each maze type and comparison setting; bolding does not indicate statistical significance.

Method	Repetition	Revision	Filler Words
Correctly Transcribed Tokens Only vs. Correctly Transcribed Tokens + Substitution Errors			
Pipeline Solution	8.46% (0.06)	4.62% (0.03)	12.84% (0.12)
E2E Solution 1	**6.14% (0.04)**	3.00% (0.02)	12.57% (0.12)
E2E Solution 2	6.99% (0.04)	**2.64% (0.01)**	**10.64% (0.09)**
Correctly Transcribed Tokens Only vs. All ASR Errors Included			
Pipeline Solution	16.70% (0.12)	12.74% (0.09)	20.06% (0.20)
E2E Solution 1	**14.95% (0.10)**	11.00% (0.07)	19.20% (0.19)
E2E Solution 2	15.41% (0.10)	**10.53% (0.07)**	**17.94% (0.16)**

Table 9. Agreement between SLP experts and automated LSA systems on LSA metrics. We report Pearson correlation, mean absolute error (MAE), mean signed difference, and standard deviation of signed differences. Values in parentheses indicate percentage differences relative to the SLP expert value. Signed difference is defined as automated system minus SLP expert, so positive values indicate overestimation. Bold values indicate the best value within each metric block (highest Pearson's *r*; smallest MAE, absolute mean signed difference, and SD of difference), with ties all bolded; bolding does not indicate statistical significance.

Metric	Method	Pearson's *r*	MAE	Mean Signed Difference (%)	SD of Difference (%)
NTU	Batchalign	0.88	16.17	14.98 (17.59%)	13.95 (14.84%)
	Pipeline Solution	**0.92**	19.88	18.03 (19.64%)	15.15 (17.61%)
	End-to-end Solution 1	0.91	17.49	14.68 (15.21%)	15.65 (19.09%)
	End-to-end Solution 2	0.90	**13.61**	**10.07 (10.35%)**	**13.84 (18.71%)**
NTW	Batchalign	0.97	53.46	38.89 (6.89%)	54.62 (9.36%)
	Pipeline Solution	**0.98**	32.78	16.89 (2.24%)	**37.31 (9.09%)**
	End-to-end Solution 1	**0.98**	**30.04**	**0.04 (-1.54%)**	43.42 (10.46%)
	End-to-end Solution 2	**0.98**	48.35	-43.74 (-10.17%)	40.26 (10.14%)
NDW	Batchalign	0.96	24.12	23.92 (16.13%)	12.51 (9.84%)
	Pipeline Solution	**0.97**	9.83	7.48 (5.03%)	**9.80 (7.85%)**
	End-to-end Solution 1	**0.97**	10.05	7.32 (4.80%)	10.84 (7.33%)
	End-to-end Solution 2	0.96	**7.72**	**2.35 (1.67%)**	**9.80 (7.55%)**
MLUm	Batchalign	**0.78**	**1.42**	**-1.36 (-18.85%)**	**0.87 (13.24%)**
	Pipeline Solution	**0.78**	1.89	-1.85 (-25.69%)	1.01 (13.29%)
	End-to-end Solution 1	0.77	1.84	-1.81 (-24.96%)	1.00 (12.92%)
	End-to-end Solution 2	**0.78**	2.10	-2.06 (-28.84%)	0.99 (13.34%)

As shown in Table 9, the automated systems exhibited complementary agreement patterns across metrics. All systems overestimated NTU, partly because the ENNI practice sections at the beginning of recordings are

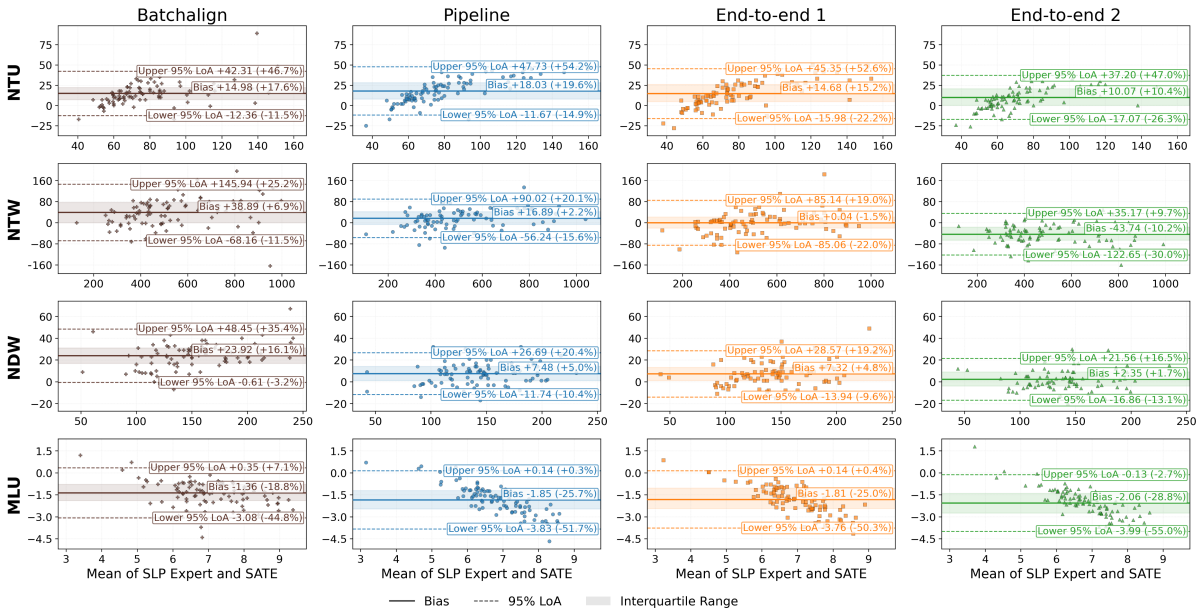


Fig. 8. Percentage Bland-Altman analysis of agreement between automated LSA systems and SLP expert-derived metrics. Rows correspond to NTU, NTW, NDW, and MLUm; columns correspond to Batchalign, Pipeline Solution, End-to-end Solution 1, and End-to-end Solution 2. Each panel reports the mean percentage bias, upper and lower 95% limits of agreement (LoA), and the interquartile range (IQR) of the percentage differences.

excluded from the ground-truth transcripts but can be transcribed by automated systems; End-to-end Solution 2 showed the smallest bias. For NTW, End-to-end Solution 1 agreed most closely with the experts, whereas Batchalign overestimated and End-to-end Solution 2 underestimated the word counts, and for NDW all SATE variants substantially reduced the positive bias of Batchalign, so the best-performing variant differs by metric. In contrast, all systems systematically underestimated MLUm, the most challenging system-level metric because it compounds errors from multiple upstream stages; MLUm estimates should therefore be interpreted cautiously and prioritized during expert review in the human-in-the-loop workflow.

Table 10 clarifies which deviations are statistically significant. Most method–metric combinations showed a significant mean percentage bias, indicating that high correlations alone should not be read as full agreement; the main exception was End-to-end Solution 1 for NTW, whose bias was not significantly different from zero. Brown-Forsythe tests found no significant differences in residual variance across systems for any metric. The differences among systems therefore lie primarily in systematic bias rather than residual variability.

Beyond numerical agreement, LSA metrics are also interpreted in terms of developmental change and group differences. As shown in Figure 12 in Appendix B, both TD and DLD groups exhibited age-related increases in most metrics in the ground-truth data, and the Pipeline Solution and both end-to-end solutions preserved these trends. The direction of the TD–DLD contrast was also largely preserved: DLD children showed lower metric values than TD children in most comparable age groups.

Taken together, automated LSA metrics are not yet interchangeable with expert-coded metrics, especially for NTU and MLUm, but SATE largely preserves the developmental trends and group-difference patterns present in

Table 10. Statistical tests of systematic percentage bias and Bland-Altman residual-variance differences. Bias p values are from one-sample t tests of whether the mean percentage difference between an automated system and SLP experts differs from zero; significant values indicate systematic over- or under-estimation, with the direction determined by the mean signed difference in Table 9. Brown-Forsythe p values test whether the variance of Bland-Altman percentage differences differs across the four automated systems for each metric; significant values would indicate different levels of residual spread across systems. Bold indicates a bias that is not significantly different from zero ($p \geq .05$), i.e., the closest agreement with experts.

Metric	Batchalign bias p	Pipeline bias p	E2E-1 bias p	E2E-2 bias p	Brown-Forsythe p
NTU	$< .001$	$< .001$	$< .001$	$< .001$	.339
NTW	$< .001$	.020	**.160**	$< .001$	.636
NDW	$< .001$	$< .001$	$< .001$	.037	.530
MLUm	$< .001$	$< .001$	$< .001$	$< .001$	.917

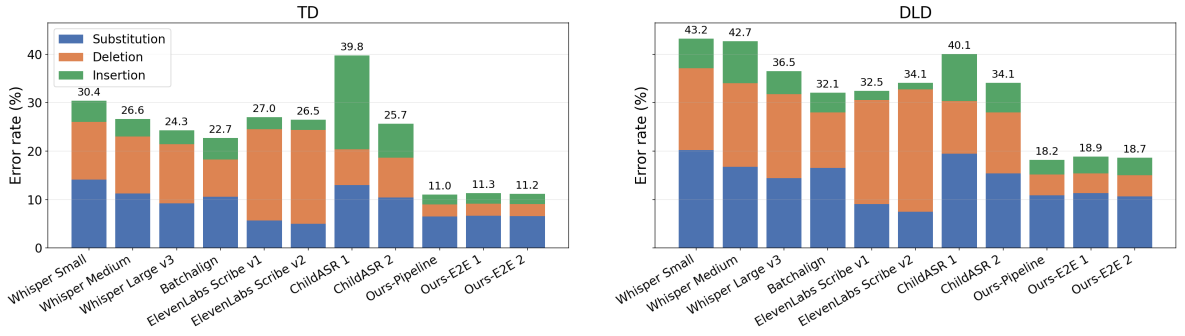


Fig. 9. WER breakdown by subgroup on the ENNI test set. Stacked bars show substitution, deletion, and insertion error rates for each ASR system, with the total WER annotated above each bar. Results are reported separately for TD and DLD children.

expert analysis. SATE therefore provides a useful automated first-pass estimate, especially for NTW and NDW, while expert verification remains essential before clinical interpretation.

5.4 Subgroup Analysis by TD/DLD Group

Because SATE is intended to support language assessment for children with DLD, we conducted an additional subgroup analysis on the held-out ENNI test set, stratified into TD children and children with DLD. We evaluated ASR performance, maze annotation performance, and agreement on LSA metrics, using the same protocols as the corresponding component-level and system-level evaluations, with full maze and LSA metric results provided in Appendix B. Because the DLD subset is considerably smaller than the TD subset (6,252 versus 51,891 reference tokens), these results are intended as descriptive subgroup evidence rather than as a powered statistical comparison between groups.

As shown in Figure 9, all systems show higher WER for DLD samples than for TD samples, indicating that DLD speech remains more challenging for automatic transcription. Nevertheless, the proposed SATE models substantially reduce WER in both groups: the pipeline ASR model achieves 11.0% WER on TD samples and

Table 11. ENNI test-set samples used in the human-in-the-loop evaluation.

Sample	Duration	C-units	Words	Maze words	Morpheme usages
A	30 s	7	59	3	7
B	27 s	7	57	2	7

18.2% on DLD samples, compared with 22.7% and 32.1% for Batchalign, and the two end-to-end variants perform comparably.

Maze annotation also remains effective for DLD samples (Appendix Table 21). Our model substantially outperforms the disfluency baseline on revision detection in both subgroups and achieves almost perfect F1 for filler words. Repetition detection shows a more nuanced pattern: our model achieves the highest F1 for TD repetitions and the highest recall for DLD repetitions, while Batchalign achieves higher DLD repetition F1 through higher precision, making repetition detection in DLD speech an important target for future improvement.

For LSA metrics (Appendix Table 22), SATE preserves strong agreement with expert analysis in both subgroups, with NTW and NDW remaining highly correlated with expert-derived values. Consistent with the overall evaluation, MLUm remains the most challenging metric, although its correlations in the DLD subgroup remain moderate to high across the three SATE variants, ranging from 0.78 to 0.87. These results support SATE's utility for children with DLD while identifying DLD speech and MLUm estimation as key targets for future robustness improvements.

5.5 Usability and Human-in-the-loop Evaluation

5.5.1 Participants and Study Protocol. To evaluate SATE as a human-in-the-loop LSA system, we designed the user study around two primary aims: (1) to evaluate whether SATE-assisted LSA reduces completion time relative to fully manual LSA, and (2) to evaluate whether SATE-assisted LSA maintains comparable accuracy after human review. The study began in February 2026. Twelve participants were recruited through a university Speech-Language Pathology (SLP) program. Most participants were SLP graduate students, and one participant was a licensed SLP. All participants had prior exposure to LSA conventions and SALT-based language sample transcription or analysis.

The study used the pipeline version of SATE, which is the configuration deployed in the current interface; the end-to-end variants were evaluated algorithmically in Section 4.2 but were not used as the source of interface outputs here. Two short language samples from the ENNI test set were selected for the main tasks, summarized in Table 11. The two samples were similar in length and linguistic complexity to support counterbalanced comparison between the SATE-assisted and manual conditions.

Each participant completed two main tasks: one SATE-assisted LSA task and one fully manual LSA task. In the SATE-assisted condition, participants uploaded the audio sample to SATE, reviewed the automatically generated pipeline output, and corrected transcription, C-unit segmentation, speaker label, maze annotation, and morpheme annotation errors when needed. In the manual condition, participants performed LSA from scratch using SALT. Task time excluded software setup. Manual timing began when the participant started playing the audio and entering the transcript in SALT and ended when the final transcript was saved. SATE timing began when the participant uploaded the audio file to SATE and ended when the corrected output was saved.

We used a within-subject counterbalanced design to control for tool order and sample differences. The 12 participants were organized into three complete groups of four participants, and within each group, participants were assigned to one of four task orders, as shown in Table 12. Thus, each group produced four SATE-assisted

Table 12. Counterbalanced task orders for the human-in-the-loop evaluation.

Order	Task 1	Task 2
1	Manual – Sample B	SATE – Sample A
2	Manual – Sample A	SATE – Sample B
3	SATE – Sample A	Manual – Sample B
4	SATE – Sample B	Manual – Sample A

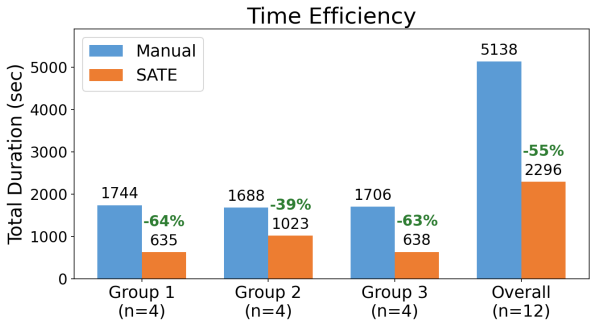


Fig. 10. Time efficiency comparison between manual LSA and SATE-assisted LSA. Each group contains four participants assigned to one complete counterbalanced set of task orders. Bars show the summed task duration within each group and overall across all 12 participants. Red labels indicate the percentage reduction in total duration when using SATE compared with manual LSA.

Table 13. Per-participant Manual and SATE durations (seconds), grouped by counterbalanced order (O1–O4).

Group	**Group 1**				**Group 2**				**Group 3**			
Participant	P1 (O1)	P2 (O2)	P3 (O3)	P4 (O4)	P5 (O1)	P6 (O2)	P7 (O3)	P8 (O4)	P9 (O1)	P10 (O2)	P11 (O3)	P12 (O4)
SATE (s)	92	118	110	315	138	395	312	178	104	121	265	148
Manual (s)	383	547	441	373	390	438	440	420	330	329	439	608

observations and four manual observations balanced across sample and task order. After completing both tasks, participants filled out a post-study questionnaire that included the System Usability Scale (SUS) [7].

5.5.2 *Results and Analysis.*

Efficiency. Efficiency was analyzed at the level of each complete counterbalanced group of four participants: within each group, we summed the four SATE-assisted and the four manual task times and compared the group-level totals, which balances the effects of task order and sample assignment.

Table 13 reports the per-participant task durations. Across all 12 participants, manual LSA required 5,138 seconds in total, whereas SATE-assisted LSA required 2,296 seconds, a reduction of 55.3%. As shown in Figure 10, SATE-assisted LSA was faster than manual LSA in all three counterbalanced groups, with group-level reductions ranging from 39.4% to 63.6%. These results provide direct evidence that the human-in-the-loop workflow substantially reduced LSA completion time while preserving expert review and correction of intermediate outputs.

Table 14. Accuracy summary for unedited SATE, SATE-assisted review, and manual LSA. For SATE-raw, values are fixed pipeline outputs for each sample. For SATE-reviewed and Manual, values are mean (SD).

	Story A			Story B		
Metric	**SATE-raw**	**SATE-reviewed**	**Manual**	**SATE-raw**	**SATE-reviewed**	**Manual**
WER	0.017	0.054 (0.045)	0.051 (0.026)	0.018	0.018 (0.000)	0.102 (0.017)
Segmentation F1	0.933	0.946 (0.048)	0.978 (0.035)	1.000	1.000 (0.000)	0.952 (0.074)
Maze Label F1	0.800	0.800 (0.000)	0.483 (0.392)	0.000	0.334 (0.365)	0.389 (0.443)
Morpheme Label F1	1.000	0.974 (0.040)	0.939 (0.054)	1.000	1.000 (0.000)	0.949 (0.040)
$\lvert\Delta \mathrm{MLU}_m\rvert$	1.125	1.007 (0.717)	0.467 (0.580)	0.143	0.072 (0.078)	0.214 (0.175)
$\lvert\Delta \mathrm{NTW}\rvert$	0.000	0.500 (0.548)	0.833 (0.753)	1.000	0.500 (0.548)	0.833 (1.169)
$\lvert\Delta \mathrm{NTU}\rvert$	1.000	0.833 (0.753)	0.333 (0.516)	0.000	0.000 (0.000)	0.000 (0.000)
$\lvert\Delta \mathrm{NDW}\rvert$	0.000	0.167 (0.408)	0.000 (0.000)	1.000	0.500 (0.548)	0.333 (0.516)

Accuracy. The accuracy analysis addressed two questions: what human review contributes to the raw pipeline output (SATE-raw versus SATE-reviewed), and how the reviewed output compares with fully manual LSA. All outputs were scored against the expert-coded reference transcripts on word error rate (WER), C-unit segmentation F1, maze label F1, morpheme label F1, and absolute errors for NTU, NTW, NDW, and MLUm. Table 14 reports the story-level results for the three conditions.

Comparing SATE-reviewed with SATE-raw in Table 14 shows that human review did not uniformly improve every metric, largely because the initial SATE outputs were already highly accurate for these short samples. Review produced targeted gains, most notably for the Story B maze labels and several final metric errors, whereas a few Story A metrics such as WER and the NTW error slightly worsened after review. This reflects the inherent ambiguity of child speech, especially for short words: different transcribers can reasonably produce slightly different transcripts, and because the ground-truth annotation is itself an expert transcription, participants' corrections represent a second subjective reading whose small deviations register as increased error, just as the fully manual condition also shows nonzero WER against the same reference. These results suggest that the value of human review was not to repair large systematic errors, but to verify the automated output and correct specific remaining errors.

We then compared SATE-reviewed with fully manual LSA within each participant using paired Wilcoxon signed-rank tests across the 12 participants, with the per-metric distributions and complete test statistics annotated in Figure 11. SATE-reviewed LSA achieved significantly lower WER ($\Delta = -0.041$, $p = 0.0073$) and significantly higher morpheme label F1 ($\Delta = +0.043$, $p = 0.0312$) than Manual LSA, while maze label F1 and segmentation F1 were numerically higher but not significantly different ($p = 0.5283$ and $p = 0.6250$, respectively). Maze label F1 was the noisiest metric, as the two reference samples contained only three and two maze words, so its null result reflects limited statistical power rather than demonstrated equivalence.

Taken together, human review left the already strong SATE-raw outputs largely unchanged while correcting specific remaining errors, and on the four transcription and annotation metrics the reviewed output was as accurate as or more accurate than fully manual LSA in this pilot study. The time savings reported above therefore did not come at the cost of analysis accuracy on these measures.

Usability. Seven of the 12 participants in the counterbalanced user study completed valid post-study questionnaires. In addition, three licensed SLPs recruited from an earlier focus group completed the same SUS questionnaire after interacting with the SATE interface. Therefore, the usability analysis included 10 valid SUS questionnaires. The three focus-group responses were included only in the usability analysis and were not used in the efficiency or accuracy analyses.

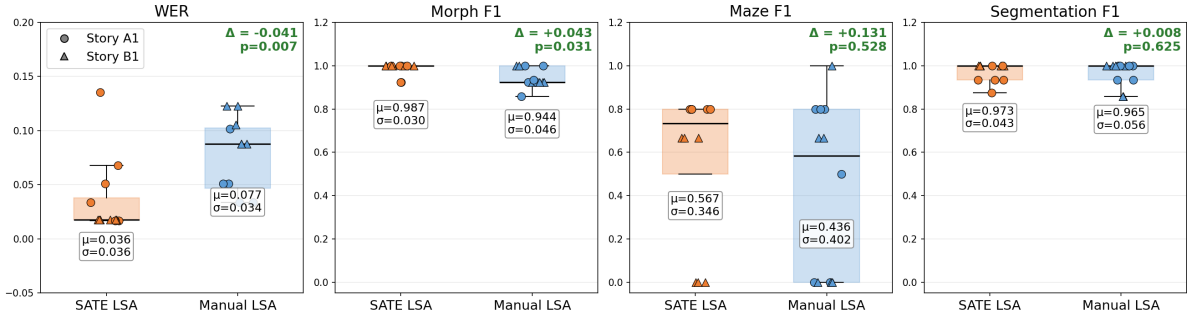


Fig. 11. Per-participant accuracy of SATE-reviewed LSA versus Manual LSA on four transcription and annotation metrics: WER, morpheme label F1, maze label F1, and segmentation F1. Box plots show the distribution across $n = 12$ participants; overlaid markers indicate individual scores, with circles for Story A and triangles for Story B. Each participant contributed one SATE-reviewed sample and one Manual sample, and story assignment was counterbalanced across tools. Within-box annotations report mean and standard deviation. Δ values report SATE − Manual, and p-values are from paired Wilcoxon signed-rank tests across participants.

Table 15. Individual SUS scores from the usability questionnaire.

Respondent	SUS	Respondent	SUS
R1	80.0	R6	92.5
R2	80.0	R7	70.0
R3	75.0	R8	72.5
R4	77.5	R9	87.5
R5	90.0	R10	87.5

The SUS is a standardized 10-item questionnaire widely used in human-computer interaction research, and has been shown to correlate with user satisfaction, learnability, and perceived efficiency across diverse systems [7]. Scores were computed using the standard procedure described in Appendix A.3. Table 15 reports the individual SUS scores, and the complete item-level responses are provided in Appendix Table 19. The mean score was 81.25 (SD = 7.75), with a 95% confidence interval of [75.71, 86.79], above the empirical SUS average of 70.14 reported across 2,324 surveys from 206 studies [3]. These results suggest that respondents perceived SATE as usable for reviewing and correcting automated LSA outputs. Given the modest sample size and the inclusion of responses from both the counterbalanced study and the earlier focus group, we interpret these findings as preliminary evidence of perceived usability rather than definitive validation of clinical deployment readiness.

In addition to perceived usability, we examined the edits participants made when reviewing SATE outputs. Edit counts were computed by comparing each participant's final SATE-reviewed output with the corresponding SATE-raw output, and thus reflect net changes in the reviewed transcript and annotations rather than low-level interaction logs. We counted four types of changes: word-level transcription, C-unit segmentation, maze label, and morpheme label edits.

As shown in Table 16, participants made only a small number of corrections: edits on Story A were mostly word-level, whereas Story B required only occasional maze label edits. This pattern suggests that SATE-raw outputs already provided a strong starting point for these short samples, and that human review primarily functioned as a targeted verification and correction step rather than a full re-annotation process.

Table 16. Edit counts made by participants on top of SATE-raw output, broken down by edit type and story. Values are mean ± SD across n=6 participants per story. A count of zero indicates that no net modification was made for that edit type.

Story	Word	Segmentation	Maze label	Morpheme label
A1 (n=6)	2.17 ± 2.64	0.50 ± 0.55	0.00 ± 0.00	0.33 ± 0.52
B1 (n=6)	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.55	0.00 ± 0.00

Beyond SUS scores and edit counts, the questionnaire included additional items about interface clarity, annotation editing, perceived accuracy, workflow fit, and remaining pain points. Appendix Table 20 summarizes this structured feedback using the exact wording of representative questionnaire statements, together with the mean rating and standard deviation.

The structured feedback highlights several strengths of the interface: all valid respondents agreed that they could quickly understand the workflow and operations of SATE, the transcription and annotation display was rated positively, and editing-related items were also positive (90% of respondents agreed that maze-label editing was intuitive and convenient, 80% that speaker-label editing was convenient, and 90% that reviewing and correcting utterances was manageable within a reasonable time). These results support the design goal of making automated LSA outputs inspectable and correctable rather than presenting them as uneditable final results.

The feedback also identified areas for improvement: trust in the final LSA metrics after manual revision was positive but moderate (mean = 3.80), ratings of automatic output quality were positive but lower than those of interface clarity and workflow learnability, and responses to the error/unexpected-behavior item were mixed, with some respondents reporting errors during use. These responses motivate clearer traceability between transcript edits, annotation changes, and metric recalculation, additional interface robustness testing, and clearer error handling.

6 DISCUSSION, LIMITATION, AND FUTURE WORK

Our experimental evaluation demonstrates that SATE advances the current LSA system across multiple dimensions. At the component level, the domain-adapted ASR model achieves 11.81% word error rate compared with 23.75% for Batchalign. The C-unit segmentation and maze annotation models similarly outperform baseline approaches when evaluated on children's speech data. Beyond these objective model performance, SATE integrates an interactive interface specifically designed for SLPs, enabling efficient review and correction of automated outputs. The human-in-the-loop evaluation shows that the interface reduced task completion time by 55.3%, supported targeted correction of automated outputs, and received a mean SUS score of 81.25, suggesting positive perceived usability. By combining high-performing automated models with human-in-the-loop verification capabilities, SATE addresses the core efficiency barrier that has limited clinical adoption of LSA.

Our comparison of pipeline and end-to-end approaches reveals complementary strengths. The pipeline solution achieves superior performance on individual components with clean inputs, offering modularity and interpretability. The end-to-end solutions demonstrate greater robustness to error accumulation, with substantially smaller performance degradation as transcription errors increase. Accordingly, we recommend the pipeline approach when transcription quality is relatively high, and the end-to-end approaches for more challenging audio conditions where higher ASR error rates are expected.

Several limitations should be acknowledged. First, all experiments were conducted on the ENNI dataset, which represents a limited sample size and a specific elicitation context. Second, agreement with SLP experts is not uniform across LSA metrics: the count-based lexical measures NTW and NDW are relatively stable, whereas NTU exhibited larger absolute differences from expert values, and MLUm showed the weakest agreement (r =

0.78), as expected for a ratio that compounds errors from multiple upstream stages; its negative mean difference indicates a tendency toward underestimation.

Clinically, the automated outputs should be treated as screening- or review-support measures rather than replacements for expert judgment: SATE captures the overall ranking and developmental patterns of children's language, but residual systematic bias can distort individual-level interpretation, for example by making a child's language appear less syntactically mature than it is and exaggerating perceived impairment severity. SATE is therefore best used as a human-in-the-loop system, in which automated estimates guide efficient review and clinicians verify C-unit boundaries, maze exclusion, and morpheme labels before making diagnostic decisions.

Future work will pursue several directions. First, rigorous clinical validation studies should examine whether SATE-assisted LSA leads to diagnostic decisions concordant with fully manual analysis while achieving significant improvements in efficiency. Second, we plan to integrate SATE with wearable recording devices to analyze children's everyday language in the settings where it is naturally produced, shifting LSA from a one-time, appointment-based assessment toward repeatable longitudinal measurement; in such continuous use, efficient human-in-the-loop verification becomes an interaction design challenge as much as an accuracy one. Third, we will extend the end-to-end architecture to jointly optimize more components such as C-unit segmentation and morphological analysis. Finally, we aim to conduct broader system deployment and data collection efforts to construct normative databases that can serve as standardized references for SLPs when making diagnostic decisions.

7 CONCLUSION

This paper presented SATE (Speech Annotation and Transcription Enhancer), an automated system addressing the efficiency barriers that have limited clinical adoption of Language Sample Analysis for child language assessment. SATE implements a machine learning pipeline reproducing the complete LSA workflow and the system is further complemented by an interactive interface supporting human-in-the-loop verification.

Our evaluation demonstrated three key findings. First, our system significantly outperforms baseline methods, achieving a word error rate of 11.81% compared with 23.75% for existing automated LSA systems on children's speech. Second, end-to-end architectures for joint transcription and maze annotation reduce error accumulation, offering improved robustness when transcription quality is degraded. Third, system-level evaluation showed strong correlations between automated and expert-produced metrics, confirming that SATE preserves diagnostically relevant patterns. A pilot human-in-the-loop evaluation provided evidence that SATE can reduce manual workload while preserving final LSA accuracy after user review.

By transforming the labor-intensive LSA workflow into an automated pipeline with transparent, verifiable, and editable intermediate representations, SATE makes gold-standard language assessment accessible at scale. For the ubiquitous computing community, SATE turns children's daily spontaneous speech into developmental measures that specialist can verify. This makes downstream applications practical: automatic analysis of each therapy session to track a child's progress during intervention, classroom deployment for large-scale DLD screening, and home deployment to study how children's language input environments shape their language development.

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A SUPPLEMENTARY USABILITY EVALUATION MATERIALS

This appendix provides supplementary materials for the human-in-the-loop usability evaluation. Section A.1 reports per-participant transcription and annotation accuracy for the two study stories. Section A.2 lists the SUS questionnaire items and complete item-level responses. Section A.3 describes the SUS scoring procedure used to compute the usability scores reported in the main text. Section A.4 reports the item-level summaries of the structured usability feedback.

A.1 Per-Participant Transcription and Annotation Accuracy

Table 17 reports the complete per-participant accuracy results for Story A1 and Story B1.

Table 17. Per-participant transcription accuracy on Story A1 and Story B1. Each cell shows "*A1 value* / *B1 value*"; the label in parentheses indicates which tool the participant used on each story. **SATE-raw** refers to the un-edited SATE pipeline output. **Metrics:** *WER*: word error rate, $(S+D+I)/N_{\text{ref}}$ over ref tokens after stripping pauses and morpheme suffixes (lower is better); *Seg F1*: C-unit boundary F1, computed by aligning ref/hyp non-maze word streams by minimum edit distance and comparing the boundary flag (last word of a C-unit) at each aligned position (TP if both flagged, FP/FN otherwise); *Maze F1*: same alignment, but comparing maze membership at each position; *Morph F1*: same alignment, but comparing the morpheme suffix (e.g. /3s, /ed) at each position; *MLU_m Diff*: hyp minus ref mean length of utterance counted in morphemes (each suffix-bearing word counts as 2); *NTW / NTU / NDW Diff*: hyp minus ref counts of total words, total C-units, and distinct word types (root forms), respectively (sign indicates direction; 0 = identical to GT).

Participant	WER	Seg F1	Maze F1	Morph F1	MLU_m Diff	NTW Diff	NTU Diff	NDW Diff
SATE-raw (A/B)	0.017/0.018	0.933/1.000	0.800/0.000	1.000/1.000	−1.125/+0.143	+0/ + 1	+1/ + 0	+0/ + 1
P1 (Manual A / SATE B)	0.102/0.018	0.933/1.000	0.000/0.000	0.923/1.000	−1.375/+0.143	−1/ + 1	+1/ + 0	+0/ + 1
P2 (SATE A / Manual B)	0.017/0.105	0.933/1.000	0.800/0.000	1.000/1.000	−1.125/−0.143	+0/ − 1	+1/ + 0	+0/ + 0
P3 (SATE A / Manual B)	0.017/0.123	0.933/0.857	0.800/0.000	1.000/0.923	−1.125/−0.143	+0/ + 0	+1/ + 0	+0/ − 1
P4 (Manual A / SATE B)	0.034/0.018	0.933/1.000	0.800/0.667	0.857/1.000	−1.000/+0.000	+1/ + 0	+1/ + 0	+0/ + 0
P5 (Manual A / SATE B)	0.034/0.018	1.000/1.000	0.800/0.667	1.000/1.000	+0.000/+0.000	+0/ + 0	+0/ + 0	+0/ + 0
P6 (SATE A / Manual B)	0.068/0.088	1.000/1.000	0.800/0.667	0.923/0.923	−0.286/−0.143	−1/ + 0	+0/ + 0	+0/ + 0
P7 (SATE A / Manual B)	0.034/0.088	1.000/1.000	0.800/0.667	0.923/0.923	−0.143/−0.143	+0/ + 0	+0/ + 0	+0/ + 0
P8 (Manual A / SATE B)	0.034/0.018	1.000/1.000	0.000/0.667	1.000/1.000	+0.286/+0.000	+2/ + 0	+0/ + 0	+0/ + 0
P9 (Manual A / SATE B)	0.051/0.018	1.000/1.000	0.500/0.000	0.923/1.000	−0.143/+0.143	+0/ + 1	+0/ + 0	+0/ + 1
P10 (SATE A / Manual B)	0.051/0.123	0.933/0.857	0.800/0.000	1.000/0.923	−1.250/−0.571	−1/ − 3	+1/ + 0	+0/ − 1
P11 (SATE A / Manual B)	0.136/0.088	0.875/1.000	0.800/1.000	1.000/1.000	−2.111/+0.143	−1/ + 1	+2/ + 0	−1/ + 0
P12 (Manual A / SATE B)	0.051/0.018	1.000/1.000	0.800/0.000	0.933/1.000	+0.000/+0.143	−1/ + 1	+0/ + 0	+0/ + 1

A.2 SUS Questionnaire and Item-Level Responses

Table 18 lists the SUS questionnaire items used in the post-study survey. Table 19 reports the complete item-level SUS responses used to compute the usability score summarized in the main usability evaluation.

A.3 SUS Scoring Procedure

SUS scores were computed following the standard scoring procedure [7]. Each of the ten SUS items was rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). For positively worded items (S1, S3, S5, S7, and S9), the item contribution was computed as the raw item score minus 1. For negatively worded items (S2, S4, S6, S8, and S10), the item contribution was computed as 5 minus the raw item score. The ten adjusted item contributions were then summed and multiplied by 2.5, yielding a total SUS score ranging from 0 to 100, where higher scores indicate better perceived usability.

Table 18. SUS questionnaire items used in the post-study survey. Negative items are reverse-scored when computing the SUS total.

Item	Question content	Scoring
S1	I think that I would like to use SATE frequently.	Positive
S2	I found SATE unnecessarily complex.	Negative
S3	I thought SATE was easy to use.	Positive
S4	I think that I would need the support of a technical person to be able to use SATE.	Negative
S5	I found the various functions in SATE were well integrated.	Positive
S6	I thought there was too much inconsistency in SATE.	Negative
S7	I would imagine that most people would learn to use SATE very quickly.	Positive
S8	I found SATE very cumbersome to use.	Negative
S9	I felt very confident using SATE.	Positive
S10	I needed to learn a lot of things before I could get going with SATE.	Negative

Table 19. Complete SUS item-level responses. S1–S10 correspond to the ten SUS items listed in Table 18. Item values are raw 1–5 Likert ratings. SUS total was computed using the standard scoring procedure.

Respondent	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	SUS
R1	5	1	4	3	4	1	4	1	3	2	80.0
R2	4	2	4	1	3	2	4	2	5	1	80.0
R3	4	2	4	2	4	2	4	2	4	2	75.0
R4	5	2	5	3	4	4	4	1	4	1	77.5
R5	5	2	5	1	4	1	5	1	3	1	90.0
R6	3	1	5	1	5	1	4	1	5	1	92.5
R7	3	2	3	1	3	1	4	3	4	2	70.0
R8	4	2	4	2	3	2	4	2	4	2	72.5
R9	3	1	4	1	5	1	5	1	4	2	87.5
R10	5	1	4	1	5	1	3	1	4	2	87.5

Formally, for respondent i, the SUS score was computed as:

$$\mathrm{SUS}_i = 2.5 \times \left(\sum_{j \in \{1,3,5,7,9\}} (x_{ij} - 1) + \sum_{j \in \{2,4,6,8,10\}} (5 - x_{ij}) \right),$$

where x_{ij} denotes respondent i's raw Likert rating for item j.

A.4 Structured Usability Feedback Items

B ADDITIONAL SUBGROUP ANALYSIS RESULTS

This appendix provides the full subgroup results for maze annotation and LSA metric agreement, together with the age-by-group distributions of LSA metrics (Figure 12). These materials support the analyses summarized in Section 5.3 and Section 5.4.

B.1 Maze Annotation Performance by TD/DLD Group

Table 21 reports the full TD/DLD subgroup maze annotation results. The same evaluation protocol and metrics are used as in the main maze annotation evaluation.

B.2 LSA Metric Agreement by TD/DLD Group

Table 22 reports the complete TD/DLD subgroup agreement results for LSA metrics.

Table 20. Structured usability feedback from questionnaire items. Representative statements are quoted directly from the questionnaire. Values indicate mean and standard deviation of valid responses on a 5-point Likert scale.

Theme	Representative questionnaire statement	Mean (SD)
Workflow learnability	"I can quickly understand the workflow and operations of SATE."	4.40 (0.52)
Interface clarity	"The transcription and annotation style are easy to understand for me."	4.40 (0.70)
Speech analysis panel	"The speech analysis panel is easy to understand for me."	4.30 (0.67)
Editing operations	"Editing maze annotations was intuitive & convenient."	4.10 (0.57)
Editing operations	"Changing speaker labels (single/bulk) was convenient."	4.10 (0.74)
Review efficiency	"Reviewing and correcting utterances was manageable within a reasonable time."	4.30 (0.67)
Automatic output quality	"Automatic maze annotations (repetition, revision, filler words, pause) were generally accurate."	4.00 (0.67)
Automatic output quality	"Automatic morpheme annotations were generally accurate."	4.10 (0.88)
Trust in final metrics	"I trust the final LSA metrics generated by SATE after my manual revisions."	3.80 (1.03)
Perceived time savings	"SATE could save me time compared to fully manual LSA."	4.78 (0.67)
Workflow fit	"SATE fits well with my current workflow."	4.57 (0.79)
Remaining pain point	"I encountered errors or unexpected behavior during the tasks."	2.60 (1.58)

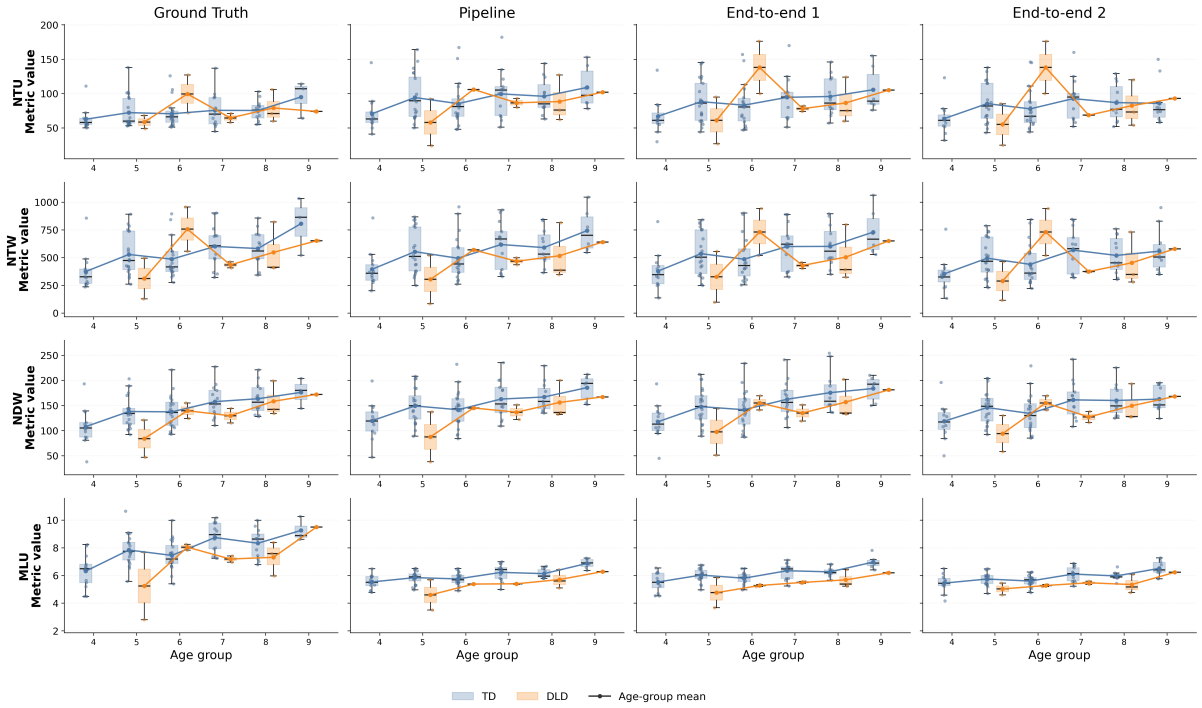


Fig. 12. Distribution of LSA metrics across age groups for TD and DLD children. Columns correspond to Ground Truth, Pipeline Solution, End-to-end Solution 1, and End-to-end Solution 2; rows correspond to NTU, NTW, NDW, and MLUm. The x-axis shows age groups from 4 to 9 years, and the y-axis shows the corresponding metric value. Within each panel, blue denotes TD and orange denotes DLD. Boxplots summarize subgroup distributions, overlaid points show individual samples, and solid lines connect age-specific subgroup means. Y-axis scales are shared within each row to facilitate comparison of the same metric across methods.

Table 21. Estimated TD/DLD subgroup maze annotation performance.

Group	Maze Type	Metric	Batchalign [22]	Disfluency [37]	Ours
TD	**Repetition**	Precision	91.17	74.20	76.40
		Recall	57.25	58.10	68.20
		F1	70.33	65.17	72.07
	Revision	Precision	N/A	72.35	77.10
		Recall	N/A	61.10	79.30
		F1	N/A	66.25	78.18
	Filler	Precision	100.00	99.50	99.84
		Recall	98.85	98.85	100.00
		F1	99.42	99.17	99.92
DLD	**Repetition**	Precision	87.82	71.60	72.85
		Recall	67.20	66.20	73.95
		F1	76.14	68.79	73.40
	Revision	Precision	N/A	66.80	72.80
		Recall	N/A	55.90	75.60
		F1	N/A	60.87	74.17
	Filler	Precision	100.00	99.05	99.10
		Recall	97.27	97.27	100.00
		F1	98.62	98.15	99.55

Table 22. Agreement between SLP experts and SATE on LSA metrics stratified by clinical group. Pearson's r, mean absolute error (MAE), mean difference, and standard deviation of difference are computed using the same protocol as the overall system-level evaluation. Values in parentheses indicate percentage differences. n denotes the number of language samples in each subgroup. DLD samples are labeled SLI in the original ENNI files.

Metric	Method	Group	n	Pearson's r	MAE	Mean Diff.	SD Diff.
NTU	Pipeline	TD	84	0.93	19.49	18.27 (20.27%)	14.71 (14.94%)
	Pipeline	DLD	8	0.81	21.63	13.13 (10.81%)	19.73 (35.69%)
	E2E-1	TD	84	0.91	16.67	14.38 (15.38%)	14.95 (17.51%)
	E2E-1	DLD	8	0.93	22.38	14.13 (10.52%)	22.44 (32.16%)
	E2E-2	TD	84	0.91	13.50	10.67 (11.36%)	13.53 (17.03%)
	E2E-2	DLD	8	0.84	14.75	3.75 (-0.27%)	16.46 (31.20%)
NTW	Pipeline	TD	84	0.98	33.06	18.85 (2.94%)	37.04 (7.84%)
	Pipeline	DLD	8	0.99	30.00	-3.75 (-5.30%)	36.19 (16.60%)
	E2E-1	TD	84	0.98	30.06	1.46 (-1.12%)	43.64 (10.25%)
	E2E-1	DLD	8	0.99	31.88	-16.88 (-6.63%)	40.28 (12.21%)
	E2E-2	TD	84	0.98	46.79	-41.74 (-9.66%)	40.21 (10.09%)
	E2E-2	DLD	8	0.99	64.75	-64.75 (-15.49%)	36.69 (9.67%)
NDW	Pipeline	TD	84	0.97	9.95	7.90 (5.44%)	9.66 (7.38%)
	Pipeline	DLD	8	0.97	8.63	4.38 (1.52%)	10.45 (11.38%)
	E2E-1	TD	84	0.96	10.06	7.27 (4.67%)	10.98 (7.36%)
	E2E-1	DLD	8	0.96	9.88	7.63 (6.17%)	10.08 (7.37%)
	E2E-2	TD	84	0.96	7.76	2.76 (1.78%)	9.86 (7.37%)
	E2E-2	DLD	8	0.99	7.25	-2.00 (0.45%)	8.49 (9.70%)
MLUm	Pipeline	TD	84	0.76	1.90	-1.86 (-25.89%)	0.99 (12.37%)
	Pipeline	DLD	8	0.87	1.75	-1.57 (-22.35%)	1.17 (20.96%)
	E2E-1	TD	84	0.77	1.83	-1.82 (-25.01%)	0.96 (11.65%)
	E2E-1	DLD	8	0.86	1.82	-1.60 (-22.97%)	1.24 (22.77%)
	E2E-2	TD	84	0.78	2.11	-2.10 (-29.53%)	0.91 (10.65%)
	E2E-2	DLD	8	0.78	2.08	-1.63 (-21.59%)	1.58 (30.07%)

C MORPHEME ANNOTATION ERROR COMPOSITION ACROSS ASR SYSTEMS

This appendix provides the error composition underlying the downstream morpheme annotation evaluation in Section 5.1.5, separating detector-induced from ASR-induced errors on both the ground-truth and prediction sides (Figure 13).

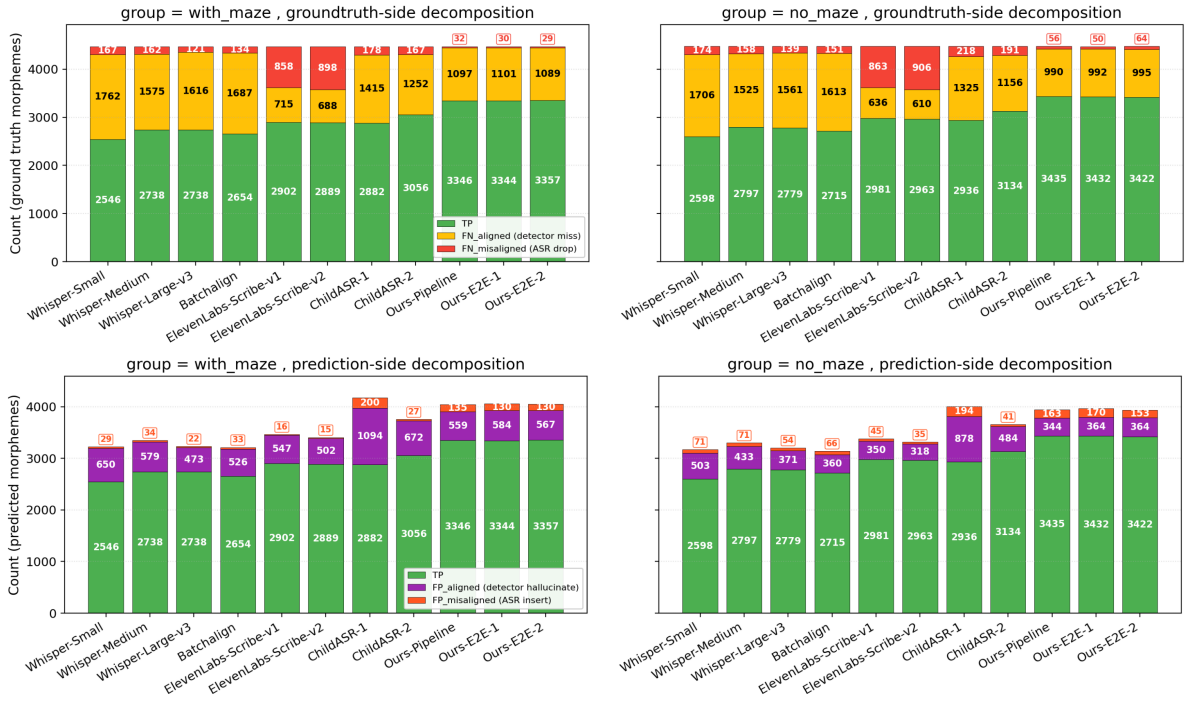


Fig. 13. Error composition of morpheme annotation across different ASR transcriptions. The first two panels decompose all ground-truth morphemes, separating TP, aligned FN, and ASR-induced misaligned FN. The last two panels decompose all predicted morphemes, separating TP, aligned FP, and ASR-induced misaligned FP. Results are shown for both original ASR transcripts with mazes retained and transcripts after maze removal.

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